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Pooling Europe's compute

The promise of distributed training
for European frontier AI

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Published by the RAND Corporation, Santa Monica, Calif., and Cambridge, UK

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Preface

Europe lags significantly behind the United States and China in frontier AI capability. The gap has multiple underlying causes, with a lack of European capital chief among them. This paper, however, focuses on a more specific challenge. To train the most advanced AI models, compute has until recently had to be concentrated at a single site, a condition particularly hard to meet in Europe given three significant bottlenecks: 1) power (slow infrastructure buildout and insufficient power at any single site); 2) compute (a small and fragmented compute stock); and 3) politics (difficulty of political coordination across member states).

Distributed training, a set of techniques allowing geographically separated datacentres to function as a single training resource, has matured rapidly since 2023. This paper assesses its relevance to European frontier AI capability, drawing on a literature review and expert interviews, as well as quantitative estimation of compute and power requirements. The analysis finds that distributed training offers relief on the power and political coordination bottlenecks. As for compute, it cannot solve the lack of chips itself, but what it does change is whether Europe's existing and planned compute can support frontier training at all: without distributed training, the same chips remain fragmented across clusters individually too small for frontier work. The report ends by setting out policy recommendations in two areas: preparing European physical infrastructure for distributed training and simplifying and harmonising cross-border regulation.

Summary

Frontier AI systems are developed primarily in the United States and increasingly in China. If Europe is to influence how AI is governed internationally, it will need sufficient domestic capability to warrant a seat at the table where the relevant rules are written and norms decided.

While Europe's lag in frontier AI has multiple underlying causes, with a lack of European capital chief among them, this paper focuses on a more specific challenge: to train the most advanced AI models, compute has until recently had to be concentrated at a single site, with hundreds of thousands of specialised chips operating in close proximity. This condition gives rise to three types of bottlenecks which stand in the way of building frontier-scale training capability in Europe: power, compute and politics. Connecting large datacentres to European grids currently takes seven to ten years, delivery times for critical equipment such as transformers and turbines have risen substantially, and no single European site currently supplies, or is on track to supply by 2030, the 5 to 10 gigawatts projected to be needed to train a frontier AI by then. Coordinating any concentrated buildout of this scale would also require political alignment across the EU's member states, a historically time-consuming endeavour.

Distributed training could offer relief: it allows datacentres in different locations to work together on the same training run, coordinated by a central entity, without needing to be physically co-located. Research progress since 2023 has substantially reduced the bandwidth between sites these techniques require, making them increasingly practical at scale. This paper examines whether distributed training can support European frontier AI capability, given the conditions shaping Europe's compute buildout, drawing on a literature review, expert interviews, and quantitative estimation of compute and power requirements.

Findings

Distributed training offers substantial relief for Europe's power and political-coordination bottlenecks but does not narrow Europe's underlying chip gap. What it does change for compute is whether the chips Europe has can be pooled into a single resource for frontier training, rather than remaining fragmented across individual clusters too small for frontier work. The following paragraphs examine each bottleneck in turn.

Power. Distributed training offers substantial relief in terms of Europe's power constraints. Utilities are typically more willing to commit power to a portfolio of smaller sites than to a single multi-gigawatt facility, where the scale of the commitment creates qualitatively different infrastructure requirements and a much larger planning risk. Connection requests for smaller sites can also proceed in parallel through different national queues, compressing overall calendar timelines compared to a single very large grid connection request. Spreading compute across grid regions furthermore allows workloads to be matched to local energy

profiles. Because the underlying algorithms for training frontier AI tolerate individual sites pausing and rejoining, distributed training may also let operators curtail load during grid peak hours without disrupting ongoing training runs, freeing additional capacity on already existing infrastructure. Distributing infrastructure across countries may also reduce concentration risk in equipment procurement and improve resilience against grid-level disruption.

Compute. Distributed training provides limited direct relief on the compute gap itself, because it does not add chips. As of the beginning of 2026, European operational capacity stands at approximately 123,000 H100-equivalents (a normalisation expressing chip throughput in units of the NVIDIA H100, the reference accelerator for frontier-training compute), against roughly 1.4 million in the United States. Including all confirmed and likely planned capacity, as well as the EU's flagship gigafactory programme, Europe reaches around 3.2 million by 2030, while the United States reaches at least 19.4 million. Several AI middle powers, including Saudi Arabia, South Korea and the United Arab Emirates, are also planning capacity that would put each ahead of Europe by 2030. Europe is therefore roughly an order of magnitude behind on both operational and planned compute, and distributed training does not change this. Distributed training does, however, determine whether the chips Europe builds can be combined into a pooled resource for more efficient frontier training, rather than remaining fragmented across clusters individually too small for frontier work.

Politics. A distributed architecture eases Europe's coordination problem by spreading the benefits of hosting AI infrastructure across more member states, including energy investment, local workforce development, and the emergence of AI-adjacent ecosystems around hosted compute. It also lowers the political threshold for engaging non-EU European countries within EFTA, particularly Norway and Switzerland, because adding new partners no longer requires negotiating their participation in a single concentrated facility. In addition, a distributed strategy can build on the existing institutional scaffolding of the EuroHPC – its procurement pipelines, its multi-sovereign decision-making processes, and its already-agreed site locations across the continent – more readily than a centralised one.

What would European distributed training look like in practice?

To illustrate what reaching frontier-competitive compute capacity in Europe would require, we set out a hypothetical scenario built on a target of 10 to 20 million H100-equivalents, the range needed to match the order of magnitude of other major and middle-power compute holders by 2030. Translating this into physical infrastructure yields a power requirement of approximately 17 to 34 gigawatts of total facility capacity. No single European site can supply this. A plausible alternative architecture would combine a small number of hyperscale clusters of roughly 2 to 3 gigawatts each, sited where large grid connections are feasible, with a larger number of smaller distributed sites spread across the continent to exploit regional energy mixes and faster connection timelines. This is one configuration among several that could in principle deliver the target, and the actual split between hyperscale and distributed capacity should emerge from a detailed siting assessment rather than from policy prescription.

Limitations of distributed training

Several limitations warrant attention. Distributed training does not address Europe's underlying capital and financial constraints, which underpin all three of the bottlenecks described above. Nor does it give Europe

a comparative advantage; the United States is already using these techniques in some of its frontier training, and China is moving in the same direction. Frontier AI developers generally prefer centralised facilities where power and grid constraints permit, and turn to multi-site configurations when forced to by external conditions. Concrete public evidence of distributed training at frontier scale remains thin, in part because frontier developers treat their training infrastructure as competitive intelligence, and because low-communication methods that may further ease training across large distances have not been demonstrated at the parameter counts of current frontier models. Furthermore, the specialised expertise required to implement distributed training at scale is concentrated almost entirely in US frontier AI companies. In addition, operating across multiple sites also introduces coordination complexity, regulatory burden and new failure modes, including dependence on inter-site network links whose disruption can halt a training run even when individual datacentres remain operational. Finally, distributed training addresses the training side of compute demand, but inference is already the dominant share of AI compute consumption and is expected to continue growing relative to training.

Recommendations

Map and prepare physical infrastructure for distributed training:

- Commission an assessment of spare grid capacity across European regions, led by the European Network of Transmission System Operators for Electricity (ENTSO-E) in collaboration with the Directorate-General for Energy (DG ENER) and member state authorities.
- Commission a parallel assessment of existing and required cross-border fibre infrastructure, led by the Directorate-General for Communications Networks, Content and Technology (DG CONNECT) in collaboration with member states and the Body of European Regulators for Electronic Communications (BEREC).
- Publish a European AI Infrastructure Reference, maintained by the European AI Office, mapping candidate datacentre sites across Europe and tagging each by suitability for different workloads.
- Reduce friction in connecting AI infrastructure to existing fibre backbones through targeted deregulation of fibre permitting and rights-of-way, and subsidies for last-mile or inter-site links where commercial provision falls short.
- Monitor trends in distributed training within the European AI Office, drawing on existing technical monitoring activities and external expertise.

Simplify and harmonise cross-border regulation:

- Accelerate and harmonise permitting and grid connection processes for AI infrastructure across member states.
- Establish a portfolio of pre-approved candidate sites with environmental assessments, zoning clearances and grid connections already in place.
- Clarify how existing data protection and sovereignty instruments apply to intra-European Union training-related data flows, through the Cloud and AI Development Act

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Abbreviations

ACER	Agency for the Cooperation of Energy Regulators
AI	Artificial Intelligence
BEREC	Body of European Regulators for Electronic Communications
BIS	Bureau of Industry and Security (US)
DG CONNECT	Directorate-General for Communications Networks, Content and Technology
DG ENER	Directorate-General for Energy
DG GROW	Directorate-General for Internal Market, Industry, Entrepreneurship and SMEs
DiLoCo	Distributed Low-Communication training
DSO	Distribution System Operator
EFTA	European Free Trade Association
ENTSO-E	European Network of Transmission System Operators for Electricity
EuroHPC	European High Performance Computing Joint Undertaking
FLAP-D	Frankfurt, London, Amsterdam, Paris, Dublin (data-centre hubs)
FLOP	Floating-Point Operation
GDP	Gross Domestic Product
GDPR	General Data Protection Regulation
GPU	Graphics Processing Unit
GW	Gigawatt
H100	NVIDIA H100 GPU (frontier-training reference accelerator)
H100e	H100-equivalent (normalised compute unit)
IEA	International Energy Agency
LLM	Large Language Model
LNG	Liquid Natural Gas
MW	Megawatt

NERC	North American Electric Reliability Corporation
PUE	Power Usage Effectiveness
TDP	Thermal Design Power
TTF	Title Transfer Facility (European wholesale gas benchmark)
VC	Venture Capital

Acknowledgements

We wish to thank Henri van Soest, James Black, Ismael Arciniegas Rueda, David Schneider-Joseph, Ben Harack, Philip Fox and Erich Grunewald for their valuable comments on drafts of this report, and the experts from across the frontier AI ecosystem who generously shared their perspectives in interviews for this study. All mistakes remain our own.

This report was supported by income from operations and gifts and grants from philanthropic supporters of the RAND Center on AI, Security, and Technology. A complete list of donors and funders is available at www.rand.org/CAST.

1. Introduction

Artificial intelligence (AI) is emerging as perhaps the most transformative general-purpose technology in history, and at this inflection point, Europe finds itself trailing behind. The technology possesses European intellectual roots, from Alan Turing’s foundational concepts of machine intelligence (Turing 1950) to Hochreiter & Schmidhuber’s (1997) development of Long Short-Term Memory networks in the 1990s, an early neural architecture for sequence modelling that preceded the transformer-based models dominating today. Yet over the course of successive AI summers and winters, the centre of gravity has shifted away from Europe. Now, in the midst of an AI boom fuelled by large language models (LLMs), the most powerful frontier AI systems are developed not in Europe, but in the United States and increasingly in China.

If Europe seeks to embed fundamental values such as human rights, the rule of law and democratic accountability into the emerging international AI order, it must have the competitive capacity to warrant a seat at the negotiating table where that order is being written. Europe does, and will, retain significant leverage through its large single market, regulatory reach and diplomatic weight (Draghi 2024). However, as AI systems increasingly determine economic productivity, defensive capability and innovative potential (Stanford HAI 2025), the actors that develop and control these systems will increasingly set the terms on which others can use them. Without reliable access to frontier AI capabilities, Europe thus risks being relegated to attempting to regulate a technology it neither builds nor fully understands.

A notable symptom of Europe’s weakness in frontier AI is the lack of compute at its disposal to support frontier AI training (Fox, Schnitzer & Privitera 2025). This is firstly a capital problem. The financing needed to build frontier-scale compute infrastructure has flowed predominantly to US and increasingly Chinese firms rather than to European ones. This capital problem has deep structural roots: shallow venture capital markets, an absence of university–venture capital talent ecosystems, and a lack of big-tech incumbents that could finance frontier capex from cash flow. These conditions in turn drive a steady brain drain of AI talent towards Silicon Valley, which further reinforces them (Draghi 2024).

A further cause relates to the set of physical conditions needed to build the infrastructure itself. Frontier AI training requires datacentre clusters drawing multiple gigawatts of power (Sevilla & Troynikov 2025). These facilities need sites with adequate grid connections, and must be built quickly enough to be commercially viable, be it by US or indigenous European hyperscalers.¹ A particular infrastructural challenge concerns running large-scale training runs on European soil. Such training runs have historically required compute

¹ Hyperscalers are large cloud infrastructure providers, such as Amazon Web Services, Microsoft Azure and Google Cloud, that design, build and operate datacentre estates at very large scale, increasingly in the multi-hundred-megawatt to gigawatt range per campus.

concentrated in a single location, because graphics processing units (GPUs) must exchange information at every training step – a level of communication that is easy to achieve inside a single datacentre but more challenging across separate sites. As a result, US companies plan to build individual datacentres the size of Manhattan (Guardian 2025). But concentrating this amount of compute at a single location requires many stars to align: sufficient power availability and grid capacity, enough space to avoid the dissent of residents, and permitting laws that allow fast building timetables. Finding suitable locations is hard enough in general, and particularly so in Europe, where existing plans for datacentre buildouts tend to be much more fragmented (European Commission 2025c).

However, a relatively new set of techniques, collectively called distributed training, allow compute to be spread across multiple sites yet function as a single large pool for training runs (Kryś, Sharma & Egan 2025). Each site performs computations locally and synchronises results at regular intervals that, depending on the algorithm used, can range from every training step to every several hundred steps (Douillard et al. 2023). This addresses a real constraint on how capacity, once built, can be used, and on the degree to which already existing infrastructure can be utilised. If distributed training proves viable at the multi-gigawatt scale required for frontier training, European compute infrastructure being spread across multiple countries from Spain to the Nordics would cease to be a disadvantage, making Europe a more viable place for frontier AI development.

Distributed training does not, of course, resolve Europe’s broader competitiveness problem, nor would it give Europe a comparative advantage: the United States is already using these techniques in some of its frontier training, and China is moving in the same direction. Facilitating large training runs is only one piece of the puzzle, and it does not substitute for ambitious economic policies to enable further datacentre buildout and attract AI talent back to the continent. But it addresses a constraint that is hard to solve through investment alone: the requirement that all compute for a frontier training run be concentrated in a single location, which even well-funded European efforts would struggle to meet given the continent’s political, energy and infrastructure situation. Realising this benefit depends on policy choices. Sufficient connectivity between datacentres is a precondition for distributed training (Sevilla & Troynikov 2025), and the regulatory, permitting and coordination frameworks needed to enable rapid distributed buildouts are not currently in place. This paper analyses to what degree the technology can be leveraged to facilitate frontier AI training on European soil, and evaluates which policies need to be in place to enable it.

Our assessment draws on a narrative literature review of recent technical papers and industry reports (2023–2025), semi-structured interviews with distributed training experts, and quantitative estimation of compute, power and timeline requirements drawing on the latest available data. Based on this analysis, we present policy options for a European frontier AI strategy employing distributed training. Section 2 describes our methodology in detail. Section 3 introduces distributed training and surveys recent technical developments. Section 4 analyses how distributed training addresses the three principal European infrastructure bottlenecks: power, compute and political coordination. Section 5 discusses cross-cutting risks and limitations. Section 6 presents policy recommendations based on our analysis.

2. Methodology

2.1. Literature review

We conducted a narrative literature review spanning multiple domains: technical papers on distributed training methods (2023–2026), industry analyses of compute scaling trends, energy infrastructure reports, and policy documents from European and international sources. A narrative rather than systematic review was chosen given the interdisciplinary scope of the research question and the need to integrate technical, infrastructural and policy sources that follow different publication conventions. For technical papers, we searched arXiv and Google Scholar using terms such as ‘distributed training’, ‘decentralized training’, ‘DiLoCo’ and ‘geo-distributed machine learning’, filtered by publication date (January 2023–April 2026). The January 2023 start date was chosen to capture the emergence of recent low-communication training algorithms, beginning with the original DiLoCo paper (Douillard et al. 2023). For energy infrastructure, we drew primarily on reports from the International Energy Agency (IEA) and the North American Electric Reliability Corporation (NERC), which provide institutional analyses of grid capacity, transmission infrastructure and supply chain constraints. These were supplemented by quantitative projections from Goldman Sachs and by trade press coverage of specific datacentre projects from outlets such as Data Center Dynamics, which we used to track site-specific developments and corporate announcements rather than as primary analytical sources. For European political and coordination dynamics, we consulted policy analyses from institutions such as the Carnegie Endowment for International Peace, the International Monetary Fund and the Delors Centre. We also monitored news coverage of European AI initiatives. Selection criteria across all domains prioritised recentness, relevance to European infrastructure context and source credibility.

2.2. Expert consultations

Between September and November 2025, we conducted four semi-structured interviews (following a flexible topic guide rather than a fixed script) with experts in distributed training research, AI infrastructure analysis and compute scaling, including researchers with published work on distributed training methods, practitioners knowledgeable in distributed and decentralised training systems, and analysts tracking frontier AI compute requirements. We cross-checked recurring expert claims against published sources where possible. In cases where these claims could be verified, we prioritised referencing published work; in cases where verification was not possible, we flag claims as expert assessments in the text.

A number of limitations apply. The sample size is small and weighed towards US-based experts, reflecting the broader concentration of distributed-training expertise in the United States (as documented elsewhere

in this paper. US-based interviewees may have limited visibility into European-specific regulatory conditions and grid interconnection speeds. In addition, some experts were working with organisations that have an interest in distributed training approaches.

2.3. Quantitative estimation

Our quantitative estimates of compute, power and timeline requirements draw on three categories of inputs. For compute, we use published estimates of recent frontier training runs, alongside projections from Epoch AI’s analytical reports and GPU Clusters dataset. Where training compute is not officially disclosed, we use the most widely cited third-party estimate and flag it as such. All compute figures are converted to H100-equivalents,² to allow comparison across hardware generations. For power, we estimate facility-level demand by combining two sources: per-GPU power figures drawn from RAND and Epoch AI, and announced or projected capacities for specific datacentre projects. Where these two methods produce similar figures, we treat the estimate as relatively robust; where they diverge, we report a range. Regional power demand projections are drawn from the IEA’s *Energy and AI* report (IEA 2025). For timelines, we compare publicly documented recent buildouts against pre-AI-boom industry benchmarks for datacentre construction. Timeline estimates are particularly uncertain because they depend on jurisdiction-specific permitting, grid interconnection queues and supply chain conditions. Where possible, we present ranges rather than point estimates, with bounds reflecting different published assumptions. We state ‘we estimate’ to flag claims that rest on our own synthesis of sources rather than a single published figure.

2.4. Definition of ‘Europe’

In this report, we focus on the potential of distributed training for Europe’s frontier AI strategy. We recognise that multiple definitions of ‘Europe’ exist, and that there are several criteria for the inclusion or non-inclusion of various countries under such definitions. We choose to limit ourselves to the European Union member states plus the four EFTA countries (Norway, Iceland, Liechtenstein and Switzerland). This makes our analysis primarily EU-focused, but also allows us to include countries that would plausibly participate in a joint European scaling effort, and many of which have the necessary infrastructure to host large numbers of data centres.

We have chosen not to include the United Kingdom in our analysis. While geographically and historically European, the UK’s AI ecosystem has diverged significantly from the EU’s since Brexit. The UK has pursued a distinct regulatory approach, declining to adopt EU-style horizontal AI regulation in favour of sector-specific guidance (DSIT 2023, 2024). Its institutional architecture has also developed along different lines.

² H100-equivalents are used throughout this report as a common unit to enable comparison across different chip generations and architectures. The measure approximates how many H100 chips, running simultaneously, would be needed to deliver the aggregate throughput (in FLOP/s) required for a given training run within its expected time frame (Epoch AI 2025; Pilz, Mahmood & Heim 2025). One important caveat is that the precision of training has shifted in recent years from mostly 16-bit to increasingly 8-bit operations, which raises effective FLOP/s on the same hardware and means that H100-equivalent counts derived from 16-bit assumptions can overstate the chip count required for an 8-bit run. Throughout this report, H100-equivalent figures should be understood as order-of-magnitude estimates rather than precise counts.

The AI Security Institute operates in close dialogue with its US counterpart within the National Institute of Standards and Technology (NIST) and has participated in bilateral rather than EU-level governance initiatives, reflecting a broader UK preference for transatlantic AI cooperation over integration with EU frameworks. The UK also maintains independent compute initiatives, such as Isambard-AI at the Bristol Centre for Supercomputing, rather than participating in joint European efforts like the EuroHPC Joint Undertaking. The UK's frontier AI activity is also structurally integrated with the US ecosystem in ways that distinguish it from continental European labs: the UK's AI scale-up ecosystem is heavily reliant on US venture capital (VC), with three large UK AI deals in 2025 alone totalling \$6.7 billion and making up about half of the country's AI VC inflows (OECD 2026). Meanwhile, the UK's most prominent frontier AI lab, Google DeepMind, operates as a subsidiary of a US parent company with its research integrated into Google's US-based product development. For these reasons, the UK's path to frontier AI capabilities, whether through distributed training or otherwise, would require a separate analysis with different assumptions about the regulatory environment, international partnerships and financing mechanisms. That said, EU pursuit of a distributed-training strategy would leave open closer future cooperation with the UK and other middle powers, including Canada, Japan, South Korea and Australia, that are increasingly coordinating on both AI capability development and AI governance.

2.5. Limitations

Several limitations apply to this paper beyond those specific to each methodological component.

Firstly, frontier AI companies treat compute capacity and infrastructure plans as competitive intelligence, and public data on frontier training runs is both limited and unevenly verified. Expert claims about corporate infrastructure plans often cannot be independently cross-checked against public sources, and we have flagged such claims in the text where they affect substantive conclusions.

Secondly, most of the technical literature on distributed training, most frontier AI corporate activity, and most of the available expert pool are concentrated in the United States, with smaller contributions from Chinese sources and a sparser set of European-specific institutional analyses. This means that claims about the US context are better supported than claims about specific European jurisdictions. Where European-specific sources exist, we draw on them directly (notably Draghi 2024, IEA 2025 and Fox, Schnitzer & Privitera 2025); where they do not, we flag the geographic asymmetry in the text.

Thirdly, this is a fast-moving field. Technical developments in distributed training, announcements of datacentre buildouts, and policy initiatives at the EU and member-state level may render some of the paper's specific estimates obsolete within months rather than years. We have attempted to distinguish between structural claims that are likely to remain valid over time and specific figures that may be superseded.

Fourthly, the paper's scope is deliberately restricted to the technical and infrastructural dimensions of European frontier AI capability and the potential contribution of distributed training therein. We do not deeply analyse adjacent questions that a comprehensive European AI strategy would need to address, including talent policy, intellectual property frameworks, semiconductor supply chains and indigenous AI company creation. This limitation defines the specific contribution of this paper rather than its completeness as a strategy document.

Fifthly, the policy recommendations in Section 6 are conditional on background assumptions about the trajectory of distributed training research, the pace of European political coordination on AI, and the broader geopolitical context in which European AI policy operates. They should be read as reasoned proposals under the conditions described in the paper, not as predictions robust to significant changes in those conditions.

3. What is distributed training and why is it useful?

The term ‘distributed training’ describes the training of AI models across multiple physically distant pools of compute, with a central entity coordinating the process (Kryś, Sharma & Egan, 2025). To evaluate the promise of distributed training for European AI ambitions, we must first understand what it is, how it differs from other methods, which bottlenecks in building AI infrastructure it addresses, and how recent developments have influenced the potential of the technology. We do not aim to cover the technical details exhaustively, but rather to provide the background needed for the policy discussion that follows. For a more detailed discussion, we recommend Kryś, Sharma & Egan (2025) for a governance-focused overview, Douillard et al. (2023) for the foundational DiLoCo (Distributed Low-Communication) framework, and Sevilla & Troynikov (2025) for an engineering analysis of distributed training at frontier scale.

3.1. What is distributed training, and what is it not?

The most common method for training frontier AI models remains centralised training, where all the GPUs used for a training run are co-located within a single datacentre or campus, connected by high-bandwidth, low-latency interconnects (Kryś, Sharma & Egan 2025). However, this definition is becoming less clear-cut; major companies such as Google increasingly use multiple datacentres within a single campus or region, connected by high-bandwidth links that effectively function as a unified facility (Gemini Team 2023). Distributed training, as discussed in this report, refers specifically to training across multiple geographically separated locations, where lower bandwidth connections necessitate different algorithmic approaches. In distributed training, the computational workload for a single model is shared across these dispersed sites while maintaining a central entity for coordination (Kryś, Sharma & Egan 2025).

Distributed training should not be confused with decentralised training, which relies on community-provided compute that is contributed on a peer-to-peer basis with no central coordinating entity (Kryś, Sharma & Egan 2025). While the two paradigms differ primarily in coordination structure rather than in any strict technical requirement, decentralised training as currently practiced depends on low-communication algorithms, since its community participants typically operate over consumer-grade internet rather than dedicated fibre.

Figure 1. Distributed training compared to adjacent techniques

		Compute locations	
		One	Many
Contributing parties	One	centralised (Grok 3)	distributed (GPT-4.5)
	Many	multi-party centralised ?	decentralised (INTELLECT-1)

Source: Kryz, Sharma & Egan 2025.

Distributed training should also not be confused with federated learning, in which training data remains local to its origin (typically mobile devices or institutional servers), and only locally computed model updates are shared with a central coordinator (McMahan et al. 2017). The motivation with this approach is to enable training on data that cannot or should not be centralised, rather than to spread compute across multiple sites.

3.2. Why is distributed training useful?

The value proposition of distributed training is that it converts geographically dispersed compute into a pooled, collectively usable resource for a single training run, thus addressing resource limitations resulting from location constraints. Hosting vast amounts of compute in close proximity requires physical infrastructure that few locations can offer.

One core constraint is the maximum power a site can draw from the grid. The power demand of a single frontier model training facility, encompassing not only the running of GPUs but also cooling, networking and other supporting infrastructure,³ has kept growing despite improvements in chip-level energy efficiency (Sevilla & Troynikov 2025). Epoch AI projects that single-campus training clusters could each demand 1 to 5 GW by 2030 (Sevilla et al. 2024), with more aggressive scenarios approaching 10 GW by the end of the decade (Sevilla & Troynikov 2025). Individual datacentre campuses in the 1–2 GW range are planned or under construction, with xAI’s Colossus currently operating at approximately 350 MW and planned to expand to 1.5 GW, and OpenAI’s Abilene Stargate operating at 240 MW with plans to reach 1.2 GW (Sevilla & Troynikov 2025). Larger multi-gigawatt projects such as OpenAI’s broader Stargate initiative (targeting 10 GW by 2029), Meta’s Hyperion (up to 5 GW) and Microsoft’s Fairwater (over 3 GW) are planned or under construction (Open AI 2025c; IEEE Spectrum 2025; Epoch AI 2025c). Scaling to 5 GW at a single site is widely considered challenging, and 10 GW clusters are likely to require the coordination of multiple datacentres across different grid regions (Sevilla & Troynikov 2025). For reference, 10 GW exceeds the nameplate capacity of the Grand Coulee Dam, the largest single power plant in the United

³ Total facility power is conventionally expressed as a Power Usage Effectiveness (PUE) multiplier on top of the power drawn by the compute hardware itself: a PUE of 1.2 means the facility uses 1.2× the IT equipment draw, with the extra 20 per cent going to cooling, networking and overhead. See Annex 1 for our assumptions.

States at 6.8 GW (Sevilla et al. 2024; Sevilla & Troynikov 2025). Distributed training would allow for the aggregation of smaller, geographically distributed datacentres, spreading out the burden between power plants.

Distributed training also addresses constraints to infrastructure buildout. Connecting a large datacentre to the grid can require waits of 2 to 10 years depending on jurisdiction, with the FLAP-D datacentre hubs (Frankfurt, London, Amsterdam, Paris, Dublin) facing average queues of 7 to 10 years (IEA 2025). Furthermore, the global delivery times for key components, such as electrical transformers and cables, have risen substantially in recent years, as surging demand from electrification and datacentre buildout has outpaced concentrated manufacturing capacity (NERC 2025; IEA 2025). Distributed training would provide an advantage here, even though it does not solve the fundamental issue of long delivery times. Spreading infrastructure across multiple sites, developers can pursue parallel buildout tracks meaning delays at one site do not halt the entire project, as other locations continue construction. Grid connection requests for smaller datacentres in different regions can also enter separate interconnection queues in parallel, avoiding the single-queue bottleneck that a large connection would face (Sevilla & Troynikov 2025). The grid connection advantage has limits, however. Hyperscale datacentres typically connect to the transmission or sub-transmission grid via dedicated on-site substations with their own large power transformers, while smaller datacentres can sometimes connect at the medium-voltage distribution level instead. The distinction matters because distribution-level connections typically require only distribution system operator (DSO) approval and minimal new infrastructure, allowing build-out in months rather than the multi-year timelines associated with transmission-level connections. Whether a smaller datacentre can connect at the medium-voltage distribution level or must still connect at transmission level depends on local utility practice and regional grid topology rather than a uniform MW threshold (GridLab 2025). In jurisdictions where distribution-level connections are available for smaller sites, the advantage is real; in others, the benefit is reduced.

A distributed approach could also address the mismatch between the energy profiles of different AI workloads and locally available power sources. Inference, the deployment phase, in which a trained model generates outputs in response to user queries, demands continuous operation with unpredictable spikes from user activity and energy requirement patterns following business hours (Chen et al. 2025), suiting regions with reliable baseload power. In contrast, training is in principle more flexible in its overall scheduling, as it can be paused and resumed, even though individual runs still produce transient power spikes during synchronisation and checkpoint events (Chen et al. 2025). Certain low-communication training algorithms introduced in Section 3.3 specifically allow individual sites to pause and rejoin without disrupting the overall run, making workload flexibility more practically achievable (Jaghoul et al. 2024a; Douillard et al. 2026). A distributed approach also allows developers to place different workloads on regions with different types of power sources. Nuclear capacity, for instance, is suitable for baseload power, while hydroelectric power plants offer flexibility to absorb demand fluctuations (IEA 2025).

In addition, a distributed approach enables a different form of flexibility: demand response. Rather than drawing continuous power from every site, distributed training allows individual sites to curtail load temporarily when the grid is stressed, because the paused sites can rejoin the training run at the next synchronisation point. The IEA estimates that if datacentres curtail load during peak demand hours, approximately 1 per cent of annual operating hours, up to 40 GW of additional datacentre capacity could

be integrated into the existing EU grid by 2035 without new transmission or generation buildout (IEA 2025, fig. 2.30). In practice, datacentre operators increasingly schedule mixed workloads to flatten power consumption, which may reduce the practical significance of workload-to-energy-source matching. However, two considerations limit how far this can go. Experts we interviewed reported that unpredictable power spikes during training runs make load-flattening considerably harder in practice than in principle. Separately, the equipment needed to smooth demand, particularly batteries and gas turbines, uses among the most supply-chain-constrained components in the current AI datacentre buildout (NERC 2025; IEA 2025). Some degree of workload-energy matching is therefore likely to remain useful, which in turn reinforces the case for placing workloads on regions whose power mix matches the workload’s needs.

Finally, distributed training also addresses political constraints on infrastructure concentration. Actors operating across multiple jurisdictions may face resistance to placing all resources in a single location. Concerns may be raised about the strategic concentration of assets or economic benefits, as well as regulatory sovereignty, particularly when a single site would host multi-gigawatt AI infrastructure. Moreover, placing all infrastructure in a single location may lead to resource conflicts with local industry, and the local population may be opposed to extreme infrastructure buildouts in their backyard – dynamics that have already driven moratoria or restrictions on hyperscale data centres in Ireland and the Netherlands (Jacamon, Dallard & Spencer 2025; Tanner et al. 2026).

3.3. Recent research advances

Before turning to recent algorithmic developments, it is worth briefly explaining the technical problem they are trying to solve. In distributed training, multiple datacentres work in parallel on different portions of the same training run, each making incremental updates to its local copy of the model. To ensure that these local copies do not drift apart, the sites must periodically synchronise, exchanging information about how each has updated the model so that all copies converge on the same final result (Duan et al. 2024). This synchronisation step is what consumes bandwidth between sites. The more frequently sites synchronise, the more bandwidth-intensive the training run becomes, and the less feasible it is to operate over slow or long-distance connections. Conventional training methods were developed for single-facility deployment, where GPUs are connected by high-speed intra-facility interconnects that make frequent synchronisation inexpensive. Applied across geographically separated sites, however, conventional methods require dedicated high-bandwidth fibre links to maintain acceptable performance. The recent algorithmic advances summarised in Table 1 therefore all share a common goal: reducing how often, or how much, sites need to communicate without degrading training quality (Douillard et al. 2023).

These advances have aimed to improve distributed training’s ability to operate across large geographic distances and with reduced communication between sites. Experts we consulted disagree, however, on how significant they are. Some argue that existing bandwidth infrastructure, including dedicated fibre, is already sufficient for distributed training across continental distances using conventional methods, and that the recent development of low-communication algorithms primarily matters for contexts where such dedicated infrastructure is unavailable. Others posit that recent advances fundamentally expand the range of feasible distributed training configurations, including those with weaker or shared network connections between sites. A separate methodological concern, raised by several experts, is that many of the recent studies rely on

simulations or bandwidth-throttled single-cluster experiments rather than real-world multi-site demonstrations, and performance has not been validated across a wide range of model scales and tasks. The table below summarises the algorithmic developments, including model scale where available, performance characteristics, and the extent to which geographic distribution has been demonstrated.

Table 1. Recent advances in multi-datacentre training

System	Scale (parameters)	Performance	Geographic distribution
DiLoCo (DeepMind, 2023)	Up to 400M	Communicates 500x less than fully synchronous training while matching its performance.	Designed for poorly connected sites but not demonstrated across large distances.
OpenDiLoco (Prime Intellect, 2024)	1.1B	Maintained 90–95 per cent efficiency on slow connections. Sites can join and leave the training run without disruption.	Real-world demonstration across two continents and three countries (Canada, Finland, US).
INTELLECT-1 (Prime Intellect, 2024)	10B	Trained on 1 trillion tokens using the PRIME framework, which combines DiLoCo with int8 quantisation to achieve a roughly 400x reduction in communication bandwidth versus standard data-parallel training. Achieved 83 per cent compute utilisation across continents and 96 per cent within a US-only configuration.	Real-world demonstration across 30 compute contributors in five countries on three continents, using up to 112 H100 GPUs simultaneously.
Streaming DiLoCo (DeepMind, 2025)	Up to 4B experimentally	Reduces required bandwidth by two orders of magnitude versus standard DiLoCo while matching training quality.	Experimental at billion-scale; geographic distribution not demonstrated.
Eager Updates (DeepMind 2025)	Up to 100B (simulated)	Allows local training to continue during synchronisation, eliminating idle time. Reaches ~95 per cent compute utilisation at 1–5 Gbit/s for models up to 100B in simulation.	Simulated only, not demonstrated at distance.
DiLoCoX (China Mobile and OG Labs, 2025)	107B	~357x speedup over standard AllReduce (the conventional bandwidth-intensive gradient synchronisation method used in centralised training) under simulated 1 Gbit/s conditions. Largest model parameter count to date among published low-communication training experiments, though achieved under bandwidth-throttled rather than genuinely distributed conditions.	Single physical cluster of 160 A800 GPUs with bandwidth artificially throttled to 1 Gbps via traffic shaping to simulate decentralised conditions. Does not demonstrate real geographic distribution.
Covenant-72B (Templar, 2026)	72B	Pre-trained on roughly 1.1 trillion tokens using SparseLoCo, a low-communication method descended from DiLoCo; performance comparable to conventionally trained models of similar scale and compute.	Real-world demonstration: trained entirely over ordinary internet connections by contributors around the world.
Decoupled DiLoCo (DeepMind, 2026)	Up to 12B	Evolution of DiLoCo which allows fully asynchronous training. Training continues even when individual sites fail or slow down, maintaining 88 per cent efficiency while matching the quality of conventional training.	Real-world demonstration across four US regions over ordinary wide-area connections (2–5 Gbit/s).

Source: Authors' compilation based on Douillard et al. (2023) for DiLoCo, Jaghouar et al. (2024) for OpenDiLoco and INTELLECT-1, Douillard et al. (2025) for Streaming DiLoCo, Kale et al. (2025) for Eager Updates, Qi et al. (2025) for DiLoCoX, Lidin et al. (2026) for Covenant-72B and Douillard et al. (2026) for Decoupled DiLoCo.

Uncertainties remain about implementation of distributed training across large distances and between multiple parties, as Table 1 shows and as we discuss in Sections 4 and 5 of this report. Real-world deployments and engineering analyses, however, suggest that the technology is already more viable than the remaining uncertainties might imply. Most frontier AI developers already use multi-datacentre training to some extent in their operations. Confirmed examples include Google's Gemini 1.0/Ultra (Gemini Team 2023) and OpenAI's GPT-4.5, which was reportedly trained across multiple campuses (OpenAI 2025a; Patel et al. 2024b). However, these training runs appear to rely primarily on standard data parallelism and pipeline parallelism, techniques that work across datacentres with sufficiently high-bandwidth connections, rather than using the low-communication methods (such as DiLoCo) discussed in Section 3.3. The Gemini 1.0 technical report, for instance, explicitly describes a data parallelism approach (Gemini Team 2023). Uncertainty remains regarding the exact configurations used by frontier companies and whether newer low-communication algorithms have been adopted at scale. That said, several of the experts we spoke to emphasised that standard training techniques can be used across datacentres in different regions without prohibitive bandwidth costs, provided that dedicated fibre links between sites are available. In essence, even without fully proven low-communication distributed training algorithms, distributed training for frontier AI development is already feasible, though primarily where high-bandwidth interconnects between sites are available. The low-communication algorithms themselves are also maturing. In April 2026, Google DeepMind reported training a 12B parameter model with its Decoupled DiLoCo framework across four geographically separate US regions over ordinary wide-area connections of 2–5 Gbit/s and described the approach as practical for production-level distributed pre-training (Douillard et al. 2026; Google DeepMind 2026). Unlike many of the studies summarised in Table 1, this was a real multi-region run rather than a simulated or bandwidth-throttled experiment, and it is the first such demonstration of these methods by a frontier AI developer. Although the demonstrated scale remains well below that of current frontier models, such advances suggest that the dependence on dedicated high-bandwidth links between sites may ease over time.

Simulations by AI trends analysis lab Epoch AI support this assessment. Sevilla & Troynikov (2025) model a 10 GW aggregate training system distributed across 23 sites spanning approximately 4,800 km in the Eastern United States (from Illinois to Philadelphia to Georgia), connected by dedicated fibre optic links. In this configuration, each site computes portions of the training workload locally and synchronises through fully synchronous data parallelism (the bandwidth-hungry synchronisation pattern used by standard data-parallel training). Although the set-up would require roughly 30 times the bandwidth of the highest-capacity transatlantic fibre cable, Epoch's simulation finds that the incremental network buildout cost is only about 0.5 per cent of total datacentre construction costs, and that the time spent synchronising would be less than 5 per cent of the time it takes to process a training batch (Sevilla & Troynikov 2025). This analysis is conservative because it assumes pure data parallelism, the most bandwidth-intensive of the conventional distributed training methods. It does not incorporate either the low-communication algorithms listed in Table 1 or more efficient conventional techniques such as pipeline parallelism, both of which would further reduce bandwidth requirements. Inter-regional distributed training may therefore be even more feasible in practice than the Epoch simulation suggests.

4. Distributed training as a potential solution to infrastructure bottlenecks

To assess the potential of distributed training for European scaling ambitions, we must first consider the bottlenecks standing in the way. Here we focus on three principal infrastructure bottlenecks that a public or private actor would face when seeking to conduct large-scale frontier AI training runs: power infrastructure, compute availability and political coordination. Sections 4.1, 4.2 and 4.3 address these bottlenecks in turn. For each, we examine how the bottleneck applies to Europe specifically, assess what relief distributed training could offer, and acknowledge the limitations of that relief. Our analysis finds that distributed training offers substantial relief for power constraints and reduces the political friction of concentrated infrastructure in a multi-state political architecture, but provides only limited direct benefit for the compute gap.

4.1. Power

4.1.1. The European power challenge

Europe's power infrastructure may stymie its AI ambitions in four ways: lengthy grid connection timelines, supply chain delays, insufficient single-site capacity, and reliability requirements that are in tension with EU energy policy.

Connecting a large datacentre to the European grid can take 7 to 10 years in the FLAP-D hubs (Frankfurt, London, Amsterdam, Paris, Dublin), compared to 1 to 3 years in the United States. This gap reflects layered permitting (EU, member-state, regional), longer environmental and consultation processes, and grid operator backlogs from renewables integration (IEA 2025). The EU has signalled its intent to accelerate this process through the proposed Cloud and AI Development Act (European Commission 2025b; 2026), which is expected to address permitting, siting and grid connection bottlenecks for AI infrastructure. But the US is starting from a substantially shorter baseline, and is already compressing its timelines further through measures such as the July 2025 executive order on accelerated federal permitting (White House 2025).

The second challenge is global rather than specifically European: energy supply chain delays. Lead times for large grid transformers reached 80 to 210 weeks (roughly 1.5 to 4 years) by 2024, approximately double their 2021 level (IEA 2025a), and have since plateaued year-on-year (NERC 2025). Building new transmission lines can take four to eight years in advanced economies (IEA 2025). Gas turbines for new gas-fired power plants now face lead times of several years, potentially delaying their commissioning beyond

2030 (IEA 2025). Battery storage systems, essential for grid reliability and demand flexibility, meanwhile face acute supply chain pressure as the expanding AI data centre buildout has placed considerable strain on the supply chain for key grid components, including low-voltage solutions and equipment supporting the integration of variable energy resources (IEA 2025). These delays are not independent. A disruption to one component cascades through procurement timelines for the others.

The third challenge is frontier AI training's rapidly growing power demand. Frontier AI companies are outlining plans for datacentres with approximately 1 GW capacity in the near term, scaling to 5 GW by 2030, with aggressive scenarios predicting 10 GW for a single geographically concentrated cluster (Sevilla & Troynikov 2025). To put that in perspective, France's peak electricity demand in 2021 was approximately 89 GW (S&P Global 2021). A single 10 GW cluster would draw roughly 11 per cent of that, in one place.

Power availability and reliability requirements create the fourth challenge, and together they generate a direct tension with EU energy policy. Meeting projected AI power demand at multi-gigawatt scale is difficult without new dispatchable generation, and Goldman Sachs projects that natural gas will provide 60 per cent of (US) expanded datacentre power demand by 2030 (Goldman Sachs 2024). On shorter timescales, during synchronisation events AI training workloads produce significant transient power spikes, which require fast-ramping generation or substantial battery storage to manage. Meeting Tier III or Tier IV reliability standards⁴ (99.982–99.995 per cent uptime) further requires layered backup infrastructure: redundant grid feeds from independent substations, uninterruptible power supplies to bridge sub-second interruptions, and on-site backup generators for extended outages. This cluster of requirements conflicts directly with the EU's REPowerEU plan, which aims to end EU dependence on Russian fossil fuels by 2027 and significantly reduce overall gas consumption by 2030 (European Commission 2022).

Taken together, these four constraints mean that any effort to build concentrated multi-gigawatt AI training infrastructure in Europe would face long timelines, constrained supply chains, insufficient single-site generation capacity, and a direct conflict with EU climate commitments. These are the conditions under which distributed training becomes most relevant, as discussed in the next subsection.

4.1.2. Distributed training as a partial solution to the power challenge

Distributed training offers partial relief on all four fronts, though it eliminates none of them outright. Specifically, it operates multiple levers, as described below.

The first lever is the scale of any single demand point. Instead of forcing one site to shoulder 5–10 GW, distributed training lets Europe spread the load across grid regions, with individual sites drawing from several hundred megawatts to a few gigawatts each. Utilities are often reluctant to sign very large power purchase agreements for multi-gigawatt single sites for several reasons. Firstly, they perceive risks that AI companies could default or that demand could fall. Secondly, serving larger loads triggers qualitatively different infrastructure requirements: at the scale of a few hundred megawatts, facilities typically require

⁴ Datacentre reliability tiers refer to the Uptime Institute's Tier Classification System, which grades facilities from Tier I (basic capacity) to Tier IV (fault-tolerant). Tier III facilities can undergo maintenance without shutdown; Tier IV facilities additionally withstand individual equipment failures without service interruption.

their own dedicated substation rather than sharing one with other customers, but at multi-gigawatt scale, entirely new transmission lines and generation capacity may be needed, introducing step-change cost increases that do not arise with smaller connections (Patel et al. 2024a; Orrick 2025; Shehabi et al. 2024). Thirdly, smaller loads can be more easily integrated into existing distribution and transmission systems without causing significant system imbalances. Finally, the failure of a single 1 GW project represents a far greater planning risk than the loss of a 100 MW site (Sevilla & Troynikov 2025).

The second lever is demand flexibility. Large datacentre operators have historically been reluctant to offer flexible load, as interruptions degrade service quality and impose throughput costs on AI training runs even where checkpointing prevents any actual loss of work. However, distributed training may change this calculus. If datacentres agree to curtail load during roughly 1 per cent of annual operating hours, the IEA estimates that up to 40 GW of additional datacentre capacity could be added to the existing EU grid by 2035 without new transmission or generation buildout (IEA 2025). Recent US analysis reaches a similar conclusion: Norris et al. (2025) find that approximately 76 GW of new US datacentre load could be accommodated on existing infrastructure if operators agree to reduce power draw during roughly 22 hours per year during seasonal grid peaks. The mechanism in both cases is the same. Datacentres draw full power for roughly 99 per cent of the year, then curtail during the handful of hours when residential, commercial and industrial demand pushes the grid to its limits. Such curtailments would allow utilities to accommodate the extra load on existing infrastructure. This flexibility is more achievable under distributed rather than centralised training. Algorithms such as DiLoCo (Douillard et al. 2023) are designed to tolerate long intervals between synchronisation steps, which means a site facing a grid peak event can pause local computation and re-join at the next synchronisation round without disrupting the other sites, a property recently demonstrated at the 10-billion-parameter scale across three continents (Jaghoul et al. 2024b).

The third lever runs through grid connection timelines. Large connection requests take substantially longer to approve than small ones (Sevilla & Troynikov 2025), so distributing the load across smaller sites cuts calendar time by putting each request into its own queue. In several European member states, connection queues for loads of several hundred megawatts currently span four to ten years, with the largest requests facing the longest delays (ENTSO-E 2024; Ember 2024). By distributing capacity across multiple smaller sites in different grid regions, Europe can pursue parallel buildout tracks that enter separate interconnection queues, potentially reducing overall calendar time compared to a single massive grid connection request.

On supply chain delays, distributed infrastructure reduces concentration risk in procurement. If delivery to one site is delayed, other sites can proceed, whereas a centralised facility faces complete delays from any single supply disruption. Distributed procurement also enables jurisdictional hedging: sourcing critical equipment across multiple supplier countries reduces exposure to any single export control regime or bilateral dispute. Partnerships with countries such as Japan, home to Mitsubishi Power, one of the three main gas turbine manufacturers globally (IEA 2025), become more strategically valuable in a distributed architecture where procurement is spread across multiple sites and countries. Such partnerships could help European companies secure priority access to critical equipment and reduce dependence on US-dominated supply chains, which, as recent analysis suggests, face significant vulnerabilities of their own.

Distributed training also aligns with Europe's heterogeneous energy mix and with the decarbonisation objectives set out in REPowerEU (European Commission 2022) and the Clean Industrial Deal (European

Commission 2025a). Whereas a centralised gigawatt-scale facility would in practice force heavy reliance on natural gas for firm capacity, distributed siting allows workloads to be matched to the energy profile of each region. For flexible training workloads, hydroelectric power in Nordic countries offers fast-ramping capacity that is well suited to interruptible training workloads and to absorbing short-term inference demand spikes. For always-on inference, nuclear infrastructure in France provides reliable low-carbon baseload power, which is particularly valuable where outages or latency increases translate directly into degraded service. Geothermal energy, while constrained by insufficient investment and early-stage technological maturity, also offers stable baseload characteristics and could support both training and inference, with Iceland representing a candidate for near-term deployment (IEA 2025). Wind and solar, while requiring substantial excess capacity and costly storage to guarantee availability, can nonetheless be leveraged through distributed siting that locates datacentres within regions of high renewable penetration, especially for flexible training jobs that can be scheduled around intermittent generation. Given their climate, Nordic countries also offer natural cooling advantages that reduce overall power consumption across both training and inference workloads. This diversity stands in contrast to a centralised approach that would require all of these characteristics at one location, likely forcing heavier reliance on gas.

Finally, geographic distribution may provide resilience against grid-level disruptions, which the IEA reports have increased fourfold between 2020 and 2024 (IEA 2025) across OECD countries (IEA 2025). A targeted disruption to energy infrastructure serving one or two datacentres could halt a centralised training run, whereas a distributed system can continue operating with degraded capacity while affected sites recover, a property that DiLoCo-style algorithms are specifically designed to support (see Section 3.4). An attack or natural disaster affecting energy infrastructure supplying one or two datacentres would more easily halt a training run than when capacity is spread across many sites. That said, it should be noted that some interviewed experts disagree with this view, pointing out that since datacentres are interconnected during distributed training, a disruption to the weakest link would be enough to halt the training.

4.1.3. Limitations and uncertainties

As demonstrated above, distributed training offers relief on energy infrastructure challenges in multiple ways. That being said, it does not fully eliminate them, and uncertainties remain. Five limitations in particular warrant attention: total power requirements are not reduced; gas dependency persists; the IEA's flexibility estimates rest on demanding preconditions; coordination complexity grows with the number of sites; and the resilience benefits of distribution are partially offset by new points of failure.

Firstly, distributed training does not reduce total power requirements. Whether Europe builds one 10 GW facility or ten 1 GW facilities, the aggregate power demand remains broadly comparable. In fact, distributed training marginally increases power demand, due to additional inter-site networking infrastructure, and performance overheads from synchronisation delays may need to be compensated with additional compute time (Jaghoul et al. 2024b). Whatever the absolute scale of capacity Europe ultimately targets, building it remains a major undertaking, and distributed training does nothing to reduce the underlying energy requirement.

Secondly, gas dependency persists. As noted in Section 4.1.1, natural gas is predicted to serve 60 per cent of US power demand for datacentres by 2030 (Goldman Sachs 2024).⁵ While hydropower can comfortably manage the predictable ramp-up and ramp-down of power demand at the start and end of training runs, Nordic hydroelectric capacity is limited in total scale relative to projected European AI energy needs (IEA 2022). It also remains unclear whether hydropower and other non-gas energy sources can adequately handle the transient power spikes that occur during training operations, sudden fluctuations driven by checkpoint saves, synchronisation events, and shifts between training phases that require rapid-response generation or storage. Battery storage is technically well-suited to absorbing such transients, but current deployment levels are insufficient and, as noted in Section 4.1.1, batteries face acute supply chain vulnerabilities (IEA 2025). This presents a challenge for a continent currently trying to heavily decrease dependence on Russian gas, substituting it for more costly alternatives such as liquid natural gas (LNG) imports from the US. The 2026 Strait of Hormuz crisis, which sharply pushed up European fuel prices on the Title Transfer Facility (TTF), Europe's main wholesale gas benchmark, has further sharpened this constraint (Tagliapietra 2026).

Thirdly, the demand flexibility claims provided by the IEA come with important caveats. The IEA notes that in order to realistically reach the added 40 GW of capacity, each site would require investment in battery storage, negotiation of throttling agreements with local grid operators, and operational systems capable of dynamically shifting workloads (IEA 2025). A parallel constraint applies on the US side: Norris et al. (2025) similarly find that the 76 GW figure depends on contractual innovation and active coordination between operators and grid authorities, neither of which is currently widespread. It remains unclear how rapidly Europe could fulfil such conditions in a coordinated manner, particularly given the heterogeneity of national permitting regimes and the limited current deployment of grid-scale battery storage.

Fourthly, experts interviewed for this report have highlighted that adding additional sites would increase coordination complexity. While parallel buildout shortens the duration of any single connection process (see Section 4.1.1), multiple grid connections mean multiple permitting processes in parallel, the navigation of heterogeneous national regulatory regimes, and thus multiple potential delays. Each datacentre location requires separate negotiations with utilities, environmental approvals, and mobilisation of construction, grid interconnection and permitting expertise, all of which are in short supply across European markets.

Finally, as pointed out by some of the experts we interviewed, the energy grid resilience benefits of distribution may be partially offset by the fact that connecting multiple datacentres creates additional points of failure. While the coordination role in distributed training can be reassigned between sites if one goes down, the inter-site network links themselves become critical infrastructure, and disruption to key connections could degrade or halt synchronisation across the network. That said, distributed training algorithms such as DiLoCo are designed to tolerate temporary site dropouts (see Section 3.4), a property demonstrated empirically at the 10-billion-parameter scale by the INTELLECT-1 run (Jaghoul et al. 2024), though tolerance to sustained or coordinated inter-site disruption at frontier scales remains untested,

⁵ While the figure indeed refers to the United States, the point that natural gas is needed for rapid spin up and spikes still carries significance in the European context.

and the severity of this risk depends on the network topology and the degree of redundancy in fibre connections between sites.

4.2. Compute

4.2.1. The European compute challenge

Compute is the second major bottleneck for European AI development. Whereas the power challenge discussed in Section 4.1 typically involves constraints caused by physical infrastructure, grid connection timelines and energy policy, the compute challenge comprises a different set of factors: access to scarce frontier chips, the capital required to procure them at scale, and the willingness of hardware suppliers and host governments to allocate cutting-edge GPUs to European buyers. This section sets out the scale of the gap between European compute capacity and that of other major actors, examines what closing the gap would require, and considers two additional challenges that make the gap harder to close: European buyers being deprioritised by chip suppliers, and shifting US export controls on frontier hardware.

The starting point for any comparison is the United States, whose compute capacity is scaling rapidly. According to Epoch AI's GPU Clusters dataset, the US held approximately 1.4 million H100 equivalents in operational clusters as of early 2026 (Epoch AI 2025a),⁶ rising to around 19.4 million when confirmed and likely planned additions are included. Europe is far behind on both counts. Based on our analysis of the same dataset, the EU, Norway and Switzerland together held approximately 122,000 H100 equivalents in operational clusters as of early 2026, rising to around 3.2 million once all confirmed and likely planned clusters are included (Epoch AI 2025a). Even Germany, Europe's largest economy, holds less than 2 per cent of global AI compute by this measure (Fox, Schnitzer & Privitera 2025). The 3.2 million figure already incorporates the EU's flagship gigafactory programme and other publicly announced European projects,⁷ several of which face significant delivery delays.

⁶ The primary data source for this analysis is Epoch AI's GPU Clusters dataset (Epoch AI 2025a), which as of January 2026 tracks 786 GPU clusters worldwide and classifies each by operational status (existing or planned) and certainty (confirmed, likely or unlikely). The dataset is biased towards publicly announced or independently verifiable clusters and likely underrepresents capacity operated by hyperscalers and frontier AI companies, which treat compute as competitive intelligence. Figures derived from it should therefore be read as lower-bound estimates. We use Epoch AI rather than other available sources because its classification system distinguishes operational from planned capacity, a distinction that several secondary sources elide.

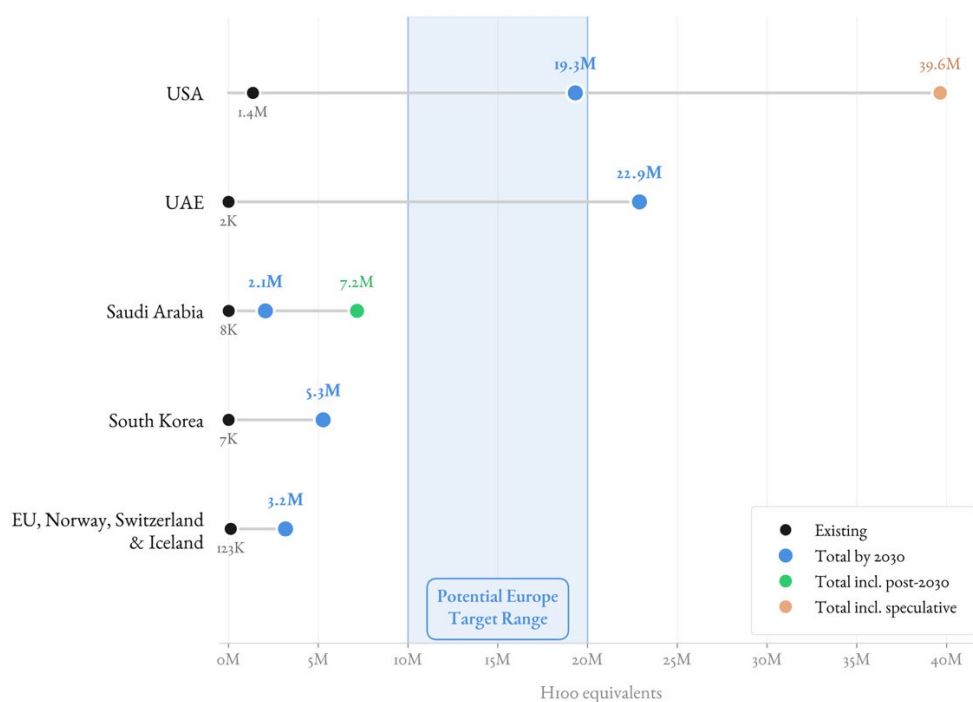
⁷ The most prominent European compute initiative is the EU's gigafactory programme, originally envisioned to deliver 500,000 GPUs across five clusters by 2026 (Reuters 2025a). The programme has experienced significant delays. While it attracted 76 expressions of interest across 16 member states and 60 proposed sites, with respondents indicating interest worth over €230 billion (European Commission 2025), the formal call for proposals was deferred from Q4 2025 to early 2026 (Light Reading 2025) and subsequently to summer 2026 (European Commission n.d.), and, and the legal basis was only established in January 2026 when Council Regulation (EU) 2026/150 entered into force (Council of the EU 2026). A second flagship project, Stargate Norway, is a joint venture between Nscale, Aker and OpenAI targeting 100,000 NVIDIA GPUs at Kvandal near Narvik (OpenAI 2025; Nscale 2025). Construction has been underway since July 2025, but the initial funded phase covers only 20 MW of the planned 230 MW capacity. Beyond these flagship initiatives, additional plans include a 1.4 GW AI datacentre campus near Paris expected by 2028 (The Register 2025) and an August 2025 EU announcement of intent to spend at least \$40 billion on US chip purchases (CEPA 2025), though the latter remains too vague to incorporate into capacity estimates.

Across the international comparison shown in Figure 2, Europe lags behind every major and middle-power compute holder. As of early 2026, the US held approximately 1.4 million H100 equivalents in operational clusters; with planned and likely capacity, this rises to around 19.4 million. The UAE, Saudi Arabia and South Korea all hold relatively small operational fleets today, but their planned capacity, 22.9 million for the UAE and 5.3 million for South Korea by 2030, and 7.2 million for Saudi Arabia by the early 2030s, places each above Europe's planned 3.2 million (Epoch AI 2025a). The Emirati figure in particular depends heavily on US chip export licensing decisions and could be substantially curtailed if those decisions go differently.

The relevant question is whether Europe can sustain a competitive pool of compute for its AI ecosystem as a whole, sufficient to support training, fine-tuning, inference and the iterative experimentation that frontier development requires across many actors simultaneously. By this measure the gap is stark.

Even a single frontier training run would exhaust Europe's planned compute base. The KIRA Center estimates that by the end of 2028, a frontier training run on the order of $1.5\text{--}2.3 \times 10^{28}$ FLOP would require 1.7 to 2.6 million H100 equivalents over a 258-day training duration (Fox, Schnitzer & Privitera 2025), taking up most or all of Europe's planned 3.2 million. By 2030, requirements rise sharply as the underlying compute scaling trend is extrapolated forward. An expert consulted for this report estimates approximately 13 million H100 equivalents being needed for a single 10^{29} FLOP run. Epoch AI's own analysis points in the same direction, calculating that 16 million H100 equivalents performing at FP8 precision and 40 per cent utilisation could complete a 2×10^{29} FLOP training run over six months (Sevilla et al. 2024). Epoch further notes that, since any single lab is likely to command only around 20 per cent of global compute production, manufacturing capacity would need to reach approximately 100 million H100 equivalents by 2030 to support frontier training of this scale.

An ecosystem capable of sustaining frontier development requires substantially more than a single training run's worth of compute. Inference at scale, fine-tuning, experimental and failed training runs, and parallel development by competing actors all draw on the same pool, and historically these uses together consume several times more compute than the headline training runs they support. For Europe to remain in roughly the same order of magnitude as other major and middle-power compute holders by 2030, an aggregate stock of around 10 to 20 million H100 equivalents would be needed, and plausibly more if Europe sought genuine self-sufficiency rather than relying on partners abroad for inference and experimentation. Against this benchmark, Europe's roughly 3.2 million planned H100 equivalents leave it approximately an order of magnitude short, behind not only the United States but also several middle powers.

Figure 2. AI compute capacity by country/region⁸

Source: Authors' calculation based on Epoch AI, Data on GPU Clusters (January 2026).

Two additional challenges make the gap harder to close. First, European companies are systematically deprioritised in the allocation of cutting-edge chips. In April 2024, Mistral CEO Arthur Mensch stated publicly that his company had access to only 1,500 H100s, a small fraction of the compute available to leading US AI companies, noting the systematic de-prioritisation of European companies by chip suppliers (Burja 2025). Second, US export policy on frontier chips has been frequently revised in recent years, and the conditions attached to chip exports increasingly extend beyond simple licensing to include detailed security and operational requirements on the receiving party, as illustrated by the elaborate conditions

⁸ Figure based on Epoch AI's Data on GPU Clusters dataset, accessed January 2026 (Epoch AI 2025a). China is excluded from the comparison due to the high opacity of Chinese figures, which makes systematic comparison unreliable. Several other caveats apply:

The figures are likely lower bounds. Some clusters classified as confirmed or likely lack H100-equivalent estimates and are excluded from the totals (e.g. the Oracle \$10 billion Saudi Arabia investment, Tesla Buffalo Dojo, IndiaAI Mission Supercomputer). Epoch AI itself estimates its dataset captures 10 to 20 per cent of global aggregate GPU performance (Epoch AI 2025b), though coverage is substantially better for the type of large publicly announced facilities relevant here, including essentially every major project from xAI Colossus to the EU AI Gigafactories. Coverage is weakest for hyperscaler custom chips (Google TPUs, AWS Trainium, Microsoft Maia), which mostly serve internal workloads rather than third-party frontier training. Separately, some likely-European clusters lack a country tag and are excluded; the most consequential is Sesterce Pegasus (~26,500 H100 equivalents, owned by the French company Sesterce), which would add less than one per cent to the European total.

The 'Total by 2030' category includes existing capacity plus confirmed and likely planned clusters dated 2030 or earlier (or undated). 'Total incl. post-2030' adds clusters dated after 2030, relevant only for Saudi Arabia, where DataVolt Neom Phase 2 (5.1 million H100 equivalents, planned for 2031) dominates planned capacity. 'Total incl. speculative' adds clusters Epoch classifies as 'unlikely', relevant only for the United States, where a rumoured Meta campus accounts for over 20 million H100 equivalents. Values are rounded to one decimal place; existing capacity below one million is shown in thousands.

attached to recent Gulf state chip import agreements (Phillips-Robins & Winter-Levy 2025). Europe has so far neither been offered nor sought such arrangements, leaving European buyers at a structural disadvantage relative to states willing to accept US oversight terms.

4.2.2. Distributed training as a partial solution to the compute challenge

Unlike with power infrastructure, distributed training offers limited direct relief for the compute bottleneck, as it does not increase the total pool of compute available. However, it does make Europe's dispersed compute stock more easily and efficiently usable for frontier training. As discussed in Sections 3 and 4.1.2, distributed training algorithms such as DiLoCo allow multiple geographically separated clusters to function as a single effective training resource by tolerating long intervals between synchronisation steps (Douillard et al. 2023; Jaghouar et al. 2024a). Recent systems can also cope with some hardware differences between sites. Decoupled DiLoCo, for example, completed a single training run on a mix of older and newer versions of Google's custom AI chips without losing performance (Douillard et al. 2026). A pooled European compute base – EuroHPC machines, AI factories and commercial clusters – would be more demanding still, since it would combine not just different chip generations but chips from different manufacturers, such as NVIDIA and AMD. Training across manufacturers has not yet been demonstrated. This matters because companies considering large-scale training runs in Europe do not just need chips to exist somewhere on the continent; they need that compute to be usable together. Without distributed training, the European compute base remains fragmented across individual clusters, none of them large enough on their own to support frontier training runs. Distributed training also builds naturally on existing European initiatives. The EuroHPC Joint Undertaking already operates supercomputer sites across multiple countries, and the AI factory and gigafactory programmes are designed to span multiple member states. This institutional and physical infrastructure makes distributed training a realistic pathway by which Europe's fragmented compute base can be combined into a single training resource at frontier scale.

Distributed training may also provide indirect benefits for chip procurement. As noted in Section 4.2.1, European companies currently sit at the back of the queue for GPU allocation from manufacturers. A distributed infrastructure strategy addresses this by spreading hosted compute across more member states. When a state hosts part of a distributed training network, it acquires a direct economic and political stake in the success of European compute procurement, which makes it more likely to back EU-level joint negotiation with chip manufacturers. It is worth noting, however, that this argument applies primarily where Europe seeks to procure chips for European-headquartered companies. US hyperscalers building European datacentres draw on their own procurement pipelines, in which they sit near the front of the queue, so the deprioritisation problem matters most where Europe seeks to build indigenous frontier capability, and matters less where the goal is hosting US-led infrastructure on European soil.

The track record on EU joint procurement is mixed. Joint vaccine procurement in 2020–21 is widely considered to have succeeded under pressure (European Commission 2021a), while the joint gas purchasing platform launched in 2022 faced structural challenges from market liberalisation and limited intra-European supply, though it has since been cited as a blueprint for future demand aggregation (Nicoli & Beetsma 2024). Chip procurement would likely face similar coordination challenges, but the political logic favouring joint action is stronger when more member states are direct stakeholders. The \$40 billion US–EU chip purchase commitment announced in September 2025, although negotiated as part of a broader

trade deal rather than as a standalone procurement initiative, illustrates that European chip transactions at this scale are now possible (Chavez 2025). Whether such commitments translate into priority access depends on the EU's collective bargaining position with NVIDIA and other suppliers, and that position is stronger when more member states have a direct stake in the outcome.

4.2.3. Limitations and uncertainties

Despite the potential benefits discussed in Section 4.2.2, distributed training does not directly close the fundamental compute gap, as it does not reduce overall GPU requirements, it provides little relief from allocation problems, it offers no protection against shifting US export controls, and the coordinated procurement it enables may slow execution.

Just as distributed training does not reduce total power requirements (Section 4.1.3), it also does not reduce overall GPU requirements. The technology changes how existing compute is used without changing how much is needed. Recent demonstrations such as INTELLECT-1 suggest the efficiency overhead can be modest at smaller scales (Jaghoul et al. 2024a), but the picture at frontier scales is less clear. Either way, the fundamental compute gap is not solved by technology.

Distributed training also offers little relief from chip allocation problems. As discussed in Section 4.2.1, European companies sit at the back of the queue for GPU allocation from manufacturers. This is a procurement and commercial relationship issue, not an infrastructure one, and rearranging where compute is physically located does not directly change which companies sit at the front of the queue.

Nor does distributed training protect against shifting US export controls. The United States has revised its export control regime on frontier AI chips multiple times since 2022, including the October 2022 Bureau of Industry and Security BIS rules, the October 2023 update, the December 2024 measures, the January 2025 AI Diffusion Rule, and the rescission of that same Diffusion Rule in May 2025 (Allen & Goldston 2025; BIS 2025). The chips inside European datacentres are subject to the same regime regardless of how those datacentres are physically configured.

Finally, while coordinated procurement may strengthen Europe's bargaining position with chip manufacturers, as discussed in Section 4.2.2, it carries an execution speed cost. EU coordination has a well-documented track record of slow decision making; the Draghi report (2024) found that EU legislation averages 19 months from Commission proposal to adoption. Distributed training therefore creates a trade-off between bargaining leverage and deployment speed. This takes us to the question of politics.

4.3. Politics

4.3.1. European coordination challenges

Europe's fragmented political authority introduces coordination challenges and conflicting economic interests that may bottleneck efforts to scale European AI compute infrastructure.

The European Union has a long history of political resistance to centralised coordination. While much of this history involves areas such as migration policy and defence, where the EU has limited formal competence, similar dynamics have emerged in adjacent economic domains where its authority is stronger: for example, Ireland, Luxembourg, Malta and Cyprus have resisted enhanced authority for the European

Securities and Markets Authority in capital markets integration (Lindner & Mack 2024), and Germany and Austria have opposed transferring banking supervisory control to supranational institutions (Balakrishnan & Pradhan, 2025). The economic stakes in AI infrastructure are large enough to intensify these dynamics. Investment in information-processing equipment and software, driven primarily by the AI datacentre buildout, accounted for an estimated 92 per cent of US gross domestic product (GDP) growth in the first half of 2025, even though this category represents only around 4 per cent of total US GDP (Armstrong & Kim 2025). Without this contribution, US GDP would have grown at just 0.1 per cent on an annualised basis. This creates powerful incentives for individual member states to compete for infrastructure placement rather than accept centralised siting decisions.

Even where member states agree on the goal of scaling European compute, regulatory and permitting timelines pose a distinct coordination challenge that depends on political will at both EU and national levels. As discussed above, pre-AI boom US datacentre buildout typically took 18 to 24 months from site selection to operation, while grid connection timelines in parts of Europe currently run to 7 to 10 years (IEA 2025). The fastest confirmed recent deployment, xAI's Colossus cluster in Memphis, reached operation in 122 days, but only by relying on contested workarounds: xAI hired 25 per cent of the US's mobile trailer cooling capacity to manage thermal loads, and used on-site gas turbines to begin operations before formal grid connection, operating in a regulatory grey area that drew permit compliance challenges (Reuters 2025a). Whether European member states possess the political will to streamline environmental permitting and grid connection to a comparable degree, and whether such approaches would survive legal challenges under existing European environmental and planning frameworks, remains uncertain.

4.3.2. Distributed training as a partial solution to coordination challenges

A distributed AI training strategy aligns with the political and economic interests of a majority of EU member states and EFTA partners. It would avoid concentrating computing power in a handful of states and would spread the associated benefits more widely (energy infrastructure upgrades, foreign direct investment, local workforce development in AI-adjacent fields, and the gradual emergence of talent and deployment ecosystems around hosted compute, as researchers, start-ups and enterprise users locate near priority-access infrastructure). It also aligns with the EU's Cohesion Policy, which aims to reduce disparities between regions and member states (European Commission 2021b). A distributed approach could extend advanced computational infrastructure to countries with limited current AI research and deployment capacity, supporting the policy's stated goal of reducing intra-EU disparities.

Non-EU EFTA partners such as Switzerland and Norway would also be easier to engage through a more distributed approach. Both countries have well-developed innovation ecosystems and significant energy resources, and Norway in particular is already host to the Stargate Norway project (OpenAI 2025; Nscale 2025). A distributed strategy lowers the political threshold for adding new partners because it does not require negotiating their participation in a single concentrated facility, making it easier to incorporate non-EU European countries into European compute initiatives.

A distributed approach also builds on existing political infrastructure. As noted in Section 4.2.2, the EuroHPC Joint Undertaking already operates a multi-sovereign governance model: 33 participating states coordinate on purchasing, deploying and operating high-performance computing systems, with supercomputers hosted in locations including Finland, Spain, Slovenia and the Czech Republic. What these

initiatives offer politically is not technical infrastructure but the institutional scaffolding around it: established procurement pipelines, multi-sovereign decision-making processes, and politically agreed site locations. These represent a foundation that a distributed scaling strategy can build on more readily than a centralised one.

4.3.3. Uncertainties and limitations

Despite this promise, a distributed approach carries its own risks and uncertainties. These range from execution challenges in government-led coordination, to the legal complexity of operating across multiple jurisdictions, to doubts about whether EU member states can agree on joint financing at the scale frontier AI infrastructure actually requires.

Government-led coordination carries real execution risks. EU coordination has a well-documented track record of extended decision-making timelines (Draghi 2024; Kirpsza 2022), and there is a real concern that heavy public involvement adds bureaucratic friction without delivering proportionate results. The AI Gigafactories programme illustrates the concern: although individual gigafactories would be smaller than some privately built datacentres completed in a matter of months, the programme remains stuck in approval processes years after announcement, with the formal call for proposals deferred repeatedly – from Q4 2025 to early 2026 and subsequently to summer 2026, still unpublished at the time of writing (European Commission 2025c, 2025d, n.d.) and the original delivery timeline no longer achievable (see Section 4.2.1). A distributed strategy only avoids this trap if the division of labour is clear: private actors should build the datacentres themselves, with governments focusing on the enabling layer of permitting, grid connections, and inter-site fibre (Kothuri et al. 2025). Where that division blurs, and where public financing or coordination is asked to do work private capital should do, distribution risks compounding the coordination burden rather than solving it.

Two different actors face regulatory burdens in a distributed architecture, and it is worth separating them. Site developers face heterogeneous permitting and planning regimes, much of which sits at the regional and municipal level rather than the national or EU one. Each datacentre site requires separate environmental review, planning approvals and compliance with local labour rules, even where EU-level frameworks exist (IEA 2025; European Commission 2025b). The parallel grid-connection-queue advantage discussed in Section 4.1.1 applies specifically to per-site connection timelines; the broader landscape of planning, environmental review and labour compliance runs in the opposite direction, with each site adding its own approval stack. AI developers running training workloads across those sites face a different layer of burden: data residency and localisation requirements that bind the training operator regardless of where the physical hardware is located. While intra-EU data transfers are protected under GDPR (Regulation (EU) 2016/679, Chapter V), pipelines that span EFTA partners or non-EU European countries face additional legal complexity, including adequacy decisions and country-specific residency requirements (EDPB 2021). We return to both issues in the policy recommendations (Section 6.2.2).

A further uncertainty sits around the financing of the public enabling layer itself. The datacentres at the heart of a distributed strategy would be privately financed, but the inter-site fibre connections, grid upgrades and coordinated permitting frameworks that make distribution viable do require public money and multi-sovereign agreement. EuroHPC, the closest existing model, operates with a 2021–2027 budget of around €7 billion (Council Regulation (EU) 2021/1173) – substantial but still an order of magnitude below the

tens of billions in public-layer financing a distributed strategy would require, and dwarfed by the hundreds of billions in private capital that would need to flow alongside it. Multi-sovereign projects also face familiar collective action problems: if each member state expects others to carry disproportionate burdens, the public component risks falling short of what is needed to unlock the private component, even where broad political support exists.

5. Cross-cutting risks and limitations of distributed training

Distributed training offers solutions to Europe’s compute infrastructure shortfall, as discussed in Sections 4.1.2, 4.2.2, and 4.3.2. It also introduces a set of risks that cut across all three domains (power, compute, politics), distinct from the domain-specific limitations covered in Sections 4.1.3, 4.2.3, and 4.3.3: operational complexity, talent requirements, and the fact that distributed training remains unproven at frontier scale.

5.1. Operational complexity and coordination costs

A distributed training strategy introduces operational complexity that cuts across all three of the bottlenecks discussed in Section 4. As detailed in Sections 4.1.3 and 4.3.3, this includes the regulatory dimension – heterogeneous national frameworks, separate utility negotiations, environmental permits, and labour-law compliance across multiple sites. A further layer, distinct from the domain-specific limitations above, is engineering complexity. Software must detect when individual sites go offline and redistribute their workloads to the remaining sites without losing training progress, tolerate higher rates of data loss between sites, and keep GPUs computing while data is being transferred between sites in order to avoid idle time on slow inter-site links. The most recent systems address several of these requirements directly. Decoupled DiLoCo, for instance, isolates site-level failures so that remaining sites continue training, maintained 88 per cent efficiency under aggressive simulated hardware failures, and allows sites to join and leave a run in progress (Douillard et al. 2026). These capabilities have, however, so far been demonstrated only within a single company's tightly integrated software stack. Multiple experts interviewed for this report identified these as serious barriers that companies would need to weigh carefully before adopting a distributed approach.

A distinct source of fragility lies in the inter-site network links themselves. In a distributed architecture, these links become critical infrastructure: disruption to the connections between sites could degrade or halt synchronisation across the whole training run, even if every individual datacentre remains fully operational. This cuts against the resilience benefit noted in Section 4.1.2, where a distributed system can continue with degraded capacity while affected sites recover. Whether distribution makes a training run more or less fragile therefore depends on the specific network topology, how many inter-site links exist, how much redundancy is built into them, and how the training algorithm handles partial network failure. Recent damage to subsea cables in the Baltic Sea, which the European Commission and High Representative have characterised as part of larger hybrid campaigns targeting EU critical infrastructure, illustrates that this is not a hypothetical concern for European cross-border infrastructure (European Commission & High Representative 2025).

The same inter-site links also expand the cybersecurity surface beyond physical disruption, creating new channels for espionage, gradient-level adversarial intervention, and exfiltration of model weights during training (Zhu, Liu & Han 2019).

Lastly, a distributed approach also sacrifices economies of scale. Centralised hyperscalers benefit from amortised infrastructure costs, bulk procurement of hardware and energy, single grid connections, and standardised operational tooling, all of which translate into substantially lower unit costs (McKinsey 2025; Pure Storage 2024).

5.2. Talent requirements

Distributed training introduces talent requirements specific to the technology. Algorithms enabling training across multiple sites and large distances draw on a specialised body of expertise, including asynchronous optimisation, gradient compression, and inter-site bandwidth-aware scheduling. This is concentrated almost exclusively in US frontier AI companies and a small number of academic groups. Even when underlying methods are published, as with DiLoCo (Douillard et al. 2023) and INTELLECT-1 (Jaghoul et al. 2024b), the implementation details, tuning and operational know-how that make them work at scale typically remain proprietary or undocumented.

Several experts interviewed for this report argued that the deeper challenge is not technical talent in isolation but the absence of an ecosystem that retains it. European researchers with relevant expertise frequently move to US labs, where the combination of compensation, compute access and career trajectory is difficult for Europe to match. Closing the talent gap therefore requires not only training programmes but also the institutional and financial conditions to keep specialised talent in Europe, whether at European firms or at local operations of foreign ones, a challenge that parallels broader European concerns about brain drain in frontier technology fields (Draghi 2024).

5.3. Unproven at frontier scale and issues in future-proofing

Distributed training has been demonstrated in smaller research trials and limited production deployments, but concrete evidence of its use by frontier AI developers at scale remains thin. Current public demonstrations involve a small number of datacentres and moderately sized models, well below the parameter counts and compute volumes characterising the latest frontier models (see Table 1). Experts caution that benchmark performance reported in research papers may obscure broader challenges in training stability, convergence quality and real-world model behaviour that emerge only at frontier scale.

Industry behaviour, while not conclusive, does offer signals. Companies prefer centralised facilities when power and grid constraints permit, and pursue distributed models mainly when forced to by external constraints (Sevilla & Troynikov 2025; Patel et al. 2024). This preference suggests that, despite recent algorithmic advances, centralised training retains significant operational and performance advantages that frontier developers thus far perceive as significant. Distributed training is, for Europe, a second-best architecture that circumstance makes hard to fully avoid. The question is not whether it is optimal but whether it is good enough to make European frontier development viable.

Centralised training is also more future-proof. Any training scheme designed for distribution can in principle adapt to centralised configurations, but communication-heavy protocols developed for centralised clusters may not translate to distributed settings. Some current AI development trends, particularly synthetic data generation and reinforcement learning fine-tuning, are relatively well-suited to distributed set-ups because they involve compute-heavy workloads with comparatively modest inter-site communication requirements (Sevilla & Troynikov 2025). Other technical advances, however, could shift this balance in either direction, and the future trajectory of training methods remains a meaningful source of uncertainty for any long-term distributed strategy.

5.4. Additional challenges

Beyond the limitations specific to distributed training, Europe faces broader structural challenges that apply to any AI competitiveness strategy.

Europe lacks both the venture capital depth and the integrated capital markets to mobilise private investment at the scale required for frontier AI infrastructure. US hyperscalers were projected in mid-2025 to deploy approximately \$350 billion in capital expenditure in 2025 and around \$400 billion in 2026 (Fusion Worldwide 2025), with individual datacentre projects increasingly in the tens of billions of dollars. More recent estimates place the figure substantially higher, with combined 2026 capital expenditure for the five largest hyperscalers (Amazon, Microsoft, Alphabet, Meta and Oracle) now forecast to exceed \$600 billion, a roughly 36 per cent year-on-year increase (IEEE ComSoc Technology Blog 2025). Europe's largest single AI infrastructure commitment to date, the gigafactory programme, totals roughly €20 billion across multiple sites (European Commission 2025e). Mistral's \$830 million debt financing for its Paris data centre buildout (CNBC 2026), on top of its €1.7 billion (\$2 billion) Series C led by ASML in September 2025 (Mistral AI 2025), represents the largest single-company financing round by a European AI company.

The Draghi report (2024) identifies a deeper structural problem: European savings rates are higher than those in the US, but the absence of an integrated capital markets union prevents these savings from being efficiently channelled into productive investment, including AI infrastructure. The result is that European private capital that could in principle finance frontier AI buildout instead flows into less productive uses or into US markets. Without addressing this structural gap, European-led infrastructure strategies, distributed or otherwise, will operate under a binding capital constraint, even where foreign-financed alternatives remain viable. This constraint cuts across each of the bottlenecks discussed in Section 4: financing power and grid buildout, procuring frontier compute, and building the joint European-level public layer that distributed coordination requires.

A further structural shift concerns the changing balance between training and inference in total AI compute demand. Distributed training addresses one side of the workload, but inference is becoming the dominant share of AI compute consumption and is expected to continue growing relative to training (Tanner et al. 2026). To the extent that the European compute challenge becomes primarily an inference challenge, the relative importance of distributed training as a policy lever will diminish accordingly.

6. Recommendations – making Europe ready for distributed training

Europe is roughly an order of magnitude behind the United States on compute, faces serious power and grid constraints, and is hampered by slow coordination. Closing this gap requires policies that address these frictions and hence incentivise compute buildout on the European continent. Distributed training has a specific role in this picture; it is the technical mechanism that allows compute spread across geographically separated sites to function as a single resource for frontier training, and could transform Europe’s fragmented compute base into a unified pool. Infrastructure to facilitate distributed training could make Europe a more attractive location for companies – both European and non-European – seeking to conduct large-scale training runs.

The primary role for European policymakers is to ensure that distributed training can be done in Europe when companies wish to do so. This requires two conditions. Firstly, mechanisms must be in place to enable rapid distributed buildout, including efficient cross-border permitting, sufficient grid capacity, and governance structures that can keep pace with private-sector deployment timelines. Secondly, infrastructure that is already built or planned should be designed from the outset to support distributed training projects, including through inter-site fibre connectivity and siting decisions that avoid unnecessary geographical isolation. Retrofitting existing infrastructure for distributed compatibility is significantly cheaper than building new sites from scratch, and European policy should make such retrofits as straightforward as possible.

The recommendations that follow build on this framing. We begin with an illustrative scenario that grounds the discussion in concrete operational terms, then turn to specific policy measures.

6.1. An illustrative scenario

Suppose Europe were to host 10 to 20 million H100 equivalents, the range identified in Section 4.2.1 as necessary to support frontier-scale training runs at a cadence sufficient for Europe to remain a credible participant in frontier AI development and regulation. What physical infrastructure and enabling

conditions would need to be in place for this compute to be usable for frontier training through distributed methods?⁹

Translating 10 to 20 million H100 equivalents into power requirements depends on assumptions about per-chip power draw, supporting hardware, and datacentre cooling efficiency. Recent estimates from RAND (Pilz, Mahmood & Heim 2025) and Epoch AI (Sevilla et al. 2024) converge on approximately 1,418 W per H100 equivalent at the cluster level, which becomes approximately 1,700 W per H100 equivalent at the total facility level under a central power usage effectiveness (PUE) assumption of 1.2 (see Annex 1). Under this assumption, 10 million H100 equivalents would require approximately 17 GW of total facility power, and 20 million approximately 34 GW.

Two sources of uncertainty complicate this central estimate. On the one hand, next-generation hardware will deliver substantially more compute per watt: for its Blackwell chips, for example, NVIDIA cites up to 25× lower energy per inference for large LLM workloads at the rack level (NVIDIA 2024). By the time European infrastructure at this scale is operational, the power requirements per unit of compute could be meaningfully lower than current estimates suggest. On the other hand, the demand-side estimates of compute required for frontier training in 2030 and beyond also rest on multiple assumptions, and given the delivery delays many projects are likely to experience, the compute requirements at the relevant time horizon could equally be higher than currently projected. Predicting how these effects will balance out is beyond the scope of this report, but it is plausible that efficiency gains will be partially or fully offset by rising compute demand. The 17 to 34 GW range should therefore be read as an order-of-magnitude working estimate rather than a precise forecast.

These power requirements are the central reason for adopting at least a partially distributed approach. No single European site, current or projected, can provide the 5 to 10 GW concentrated at one cluster that a fully centralised approach would require: even the largest planned European facility, the 1.4 GW Campus AI project in the Paris region (École Polytechnique 2025), falls well below this threshold. Grid connections at the required scale face approval timelines of 7 to 10 years in many European countries (IEA 2025), and a fully centralised approach would forgo the demand-flexibility benefits discussed in Section 4.1.2, which could unlock tens of gigawatts of additional capacity using existing infrastructure. Smaller sites in the 300 MW to 1 GW range can connect to grids substantially faster, and parallel buildout across multiple locations would compress the overall timeline considerably (Sevilla & Troynikov 2025; NERC 2025). A fully centralised approach would also face political resistance from medium and small EU member states and would risk excluding valuable partners such as Switzerland and Norway, as discussed in Section 4.3.2.

⁹ The technical foundations for European-scale distributed training are more mature than is often assumed. Frontier AI companies already train across multiple datacentres (see Section 3.3), and Epoch AI's simulation of a 10 GW system distributed across 23 US sites estimated performance overhead below 5 per cent relative to an equivalent centralised configuration (Sevilla & Troynikov 2025). Several of the experts consulted for this report assessed existing methods as sufficient for inter-regional training where high-bandwidth interconnect is available between sites, suggesting that distributed training at this scale is moving beyond purely theoretical territory. Two qualifications are worth noting. Firstly, the geographic distances involved in European distributed training would substantially exceed those in the Epoch AI simulation, and the performance overhead at intra-European scales has not yet been empirically validated. Secondly, the expert assessment is conditional on inter-site bandwidth that does not yet exist across most European links and would itself require substantial investment.

One plausible set-up would involve two tracks. The first would consist of 4 to 6 hyperscale clusters of approximately 2 to 3 GW each, located where large-scale grid connections are most feasible, for example near France’s nuclear baseload or Nordic hydroelectric infrastructure. These provide the concentrated compute density needed for the most bandwidth-intensive training phases and for large-scale inference.

The second track would complement these with 15 to 25 distributed sites of 500 MW to 1 GW each, exploiting demand flexibility, faster grid connections, and Europe’s varied energy landscape (for example Iberian solar, North Sea wind, or geothermal sources in Iceland and Italy). Combined, the two tracks would provide approximately 16 to 43 GW of total facility power, bracketing the 17 to 34 GW range implied by the 10 to 20 million H100 equivalent central estimate.

During a frontier training run, the hyperscale clusters would handle the most communication-intensive portions of the workload, while the distributed sites contribute additional compute capacity through low-communication algorithms such as DiLoCo (Douillard et al. 2023), synchronising at intervals rather than continuously. Between major training runs, both tracks would serve inference, fine-tuning and experimental workloads, with the distributed sites providing demand flexibility by temporarily reducing load during grid peaks while the hyperscale clusters maintain baseline operations. The precise allocation of workloads across the two tracks would depend on bandwidth, training architecture and relative cost at each location, with these decisions best left to operators rather than prescribed by policy. The scenario shows that a 17 to 34 GW European compute base is constructible by pairing a small number of hyperscale anchor sites with a fleet of smaller distributed clusters, rather than concentrating frontier training into a single ultra-large facility. The binding constraints are therefore primarily regulatory and infrastructural rather than technological, which is what the policy recommendations below address.

Several caveats apply. This is one illustrative scenario, not a blueprint. Investment in energy supply, semiconductors and talent remains equally critical. The power conversion reflects current-generation hardware and will shift substantially as more efficient chips reach deployment. The dual-track architecture assumes continued maturation of distributed training algorithms, which remain largely unproven at full frontier scale. The precise split between hyperscale and distributed capacity should be determined by a detailed siting assessment, which we recommend commissioning as an early priority.

6.2. Policy recommendations

The following policy recommendations focus on helping to create a landscape that is prepared for the inclusion of a distributed training strategy into Europe’s frontier AI scaling. The recommendations centre around two pillars: mapping and preparing physical infrastructure for distributed training, and simplifying and harmonising cross-border regulation.

6.2.1. Map and prepare physical infrastructure for distributed training

Enabling distributed training in Europe requires the right physical preconditions. This means mapping current infrastructure, identifying gaps, designing new infrastructure with distributed training in mind from the outset, as well as tracking technical developments closely enough that policy does not lag behind the state of the art.

Assess spare energy capacity across European grid regions. This assessment could be led by ENTSO-E in collaboration with DG ENER and member state authorities. The purpose would be to map facilities that operate below their grid allocations, regions with available headroom, and grid connection timelines by jurisdiction. The assessment should also identify where flexibility in demand from operators could unlock additional capacity.

Assess existing and required cross-border fibre infrastructure. This assessment could be led by DG CONNECT in collaboration with member state authorities and the Body of European Regulators for Electronic Communications (BEREC), and it should map both existing fibre capacity and the gaps that would need to be filled to enable a pan-European distributed training architecture.

Publish a European AI Infrastructure Reference. Based on energy capacity and fibre assessments, the EU AI Office should produce and maintain a public database that contains two components: firstly, a map of candidate datacentre sites across Europe above 100MW, with each location tagged for the type of workload it best supports (such as distributed training, hyperscale baseload, inference and so on); secondly, a query function that allows users to filter locations by their own criteria, such as available power capacity, energy mix and latency to other sites. The reference will serve as a guidance document for companies evaluating Europe for frontier AI infrastructure buildouts, and as a resource for EU and member state institutions looking to prioritise investment into enabling infrastructure for datacentres.

Make AI infrastructure easy to interconnect across Europe. EU and member state authorities should reduce the friction of connecting AI infrastructure to existing fibre backbones through targeted deregulation of fibre permitting and rights-of-way, and subsidies for last-mile or inter-site links where commercial provision falls short. Co-financing institutions such as the European Investment Bank (EIB) and EuroHPC can reinforce this by favouring projects sited near existing fibre backbones, without asking operators to build fibre themselves.

Monitor trends in distributed training within the EU AI Office. Distributed training is a fast-developing technical domain, and as a result associated policy recommendations may quickly become out of date. To address this moving target problem, the EU AI Office should incorporate distributed training into its existing technical monitoring activities, drawing on external expertise where needed rather than creating new dedicated capacity.

6.2.2. Simplify and harmonise cross-border regulation

Companies seeking to operate AI training infrastructure across Europe currently face patchy and slow regulatory regimes and processes, making it unattractive to develop frontier AI models via distributed training across European countries. Reducing this burden is valuable regardless of whether distributed training ends up being used as a main mechanism for training or not. The following actions are recommended:

Accelerate and harmonise permitting and grid connection processes for AI infrastructure across the continent. DG GROW and DG ENER, in cooperation with member states and the Agency for the Cooperation of Energy Regulators (ACER), should ensure that an AI company can go from investment decision to construction seamlessly, so that datacentres can go online in internationally competitive timelines.

Establish a portfolio of pre-approved candidate sites. DG ENER, DG GROW and the EU AI Office should create ‘pre-approved’ AI infrastructure sites, where grid connections, environmental assessments, zoning clearances and other documentation needs are accelerated. Member state authorities and energy operators could then execute actual site preparation within their jurisdictions. The EU AI Office should maintain a publicly available database of these sites.

Resolve the legal issues of operating data flows between training sites. During distributed training, a high volume of data crosses between datacentres. The legal status of these flows under existing data protection rules (notably the General Data Protection Regulation) and sovereignty instruments (including the Data Act, the EUCS scheme under the Cybersecurity Act, and the Cloud and AI Development Act) needs to become clearer. DG CONNECT should use the Cloud and AI Development Act, proposed by the Commission in June 2026 (European Commission 2026), to clarify how these instruments apply to intra-EU data flows arising from distributed model training, so that companies can run cross-border training runs within Europe with legal certainty.

References

- Allen, G.C., & I. Goldston. 2025. *Understanding U.S. allies' current legal authority to implement AI and semiconductor export controls*. Center for Strategic and International Studies, 14 March. As of 30 June 2026:
<https://www.csis.org/analysis/understanding-us-allies-current-legal-authority-implement-ai-and-semiconductor-export>
- Armstrong, R., & H. Kim. 2025. 'Does GDP growth minus AI capex equal zero?' *Financial Times*, Unhedged, 1 October. As of 30 June 2026:
<https://www.ft.com/content/603be7d5-13ac-4047-9854-24db5f0f0802>
- Balakrishnan, R., & M. Pradhan. 2025. 'Europe's elusive savings and investment union.' International Monetary Fund, Finance & Development, June. As of 30 June 2026:
<https://www.imf.org/en/publications/fandd/issues/2025/06/europes-elusive-savings-and-investment-union-ravi-balakrishnan>
- BIS (Bureau of Industry and Security). 2025. 'Department of Commerce announces rescission of Biden-era Artificial Intelligence Diffusion Rule.' US Department of Commerce, 13 May. As of 30 June 2026:
<https://www.bis.gov/press-release/department-commerce-announces-rescission-biden-era-artificial-intelligence-diffusion-rule-strengthens>
- Burja, S. 2025. 'AI 2026: Mistral will rise as compute is unleashed.' Bismarck Brief, 10 September. As of 30 June 2026: <https://brief.bismarckanalysis.com/p/ai-2026-mistral-will-rise-as-compute>
- Chavez, P. 2025. 'What the US-EU \$40 billion chip deal means.' Center for European Policy Analysis, 2 September. As of 30 June 2026:
<https://cepa.org/article/what-the-us-eu-40-billion-chip-deal-means/>
- Chen, X., X. Wang, A. Colacelli, M. Lee & L. Xie. 2025. 'Electricity demand and grid impacts of AI data centers: Challenges and prospects.' arXiv:2509.07218. As of 30 June 2026:
<https://arxiv.org/abs/2509.07218>
- CNBC. 2026. 'Mistral secures \$830 million in debt financing to fund AI data center.' 30 March. As of 30 June 2026:
<https://www.cnbc.com/2026/03/30/mistral-ai-paris-data-center-cluster-debt-financing.html>

- Council of the European Union. 2026. 'Artificial intelligence: Council paves the way for the creation of AI gigafactories.' Press release, 16 January. As of 30 June 2026:
<https://www.consilium.europa.eu/en/press/press-releases/2026/01/16/artificial-intelligence-council-paves-the-way-for-the-creation-of-ai-gigafactories/>
- Douillard, A., Q. Feng, A.A. Rusu, R. Chhaparia, Y. Donchev, A. Kuncoro, et al. 2023. 'DiLoCo: Distributed low-communication training of language models.' arXiv:2311.08105. As of 30 June 2026: <https://arxiv.org/abs/2311.08105>
- Douillard, A., K. Rush, Y. Donchev, Z. Charles, A. Dubey, B. Woodworth, et al. 2026. 'Decoupled DiLoCo for resilient distributed pre-training.' arXiv:2604.21428. As of 8 July 2026:
<https://arxiv.org/abs/2604.21428>
- Douillard, A., Y. Donchev, K. Rush, S. Kale, Z. Charles, Z. Garrett, et al. 2025. 'Streaming DiLoCo with overlapping communication: Towards a distributed free lunch.' arXiv:2501.18512. As of 30 June 2026: <https://arxiv.org/abs/2501.18512>
- Draghi, M. 2024. *The future of European competitiveness*. Report to the European Commission, Brussels, September. As of 30 June 2026:
https://commission.europa.eu/topics/eu-competitiveness/draghi-report_en
- DSIT (UK Department for Science, Innovation and Technology). 2023. *A pro-innovation approach to AI regulation*. White Paper CP 815, 29 March. As of 30 June 2026:
<https://www.gov.uk/government/publications/ai-regulation-a-pro-innovation-approach>
- . 2024. *A pro-innovation approach to AI regulation: Government response*. Command Paper 1019, 6 February. As of 30 June 2026:
<https://www.gov.uk/government/consultations/ai-regulation-a-pro-innovation-approach-policy-proposals/outcome/a-pro-innovation-approach-to-ai-regulation-government-response>
- Duan, J., S. Zhang, Z. Wang, L. Jiang, W. Qu, Q. Hu, et al. 2024. 'Efficient training of large language models on distributed infrastructures: A survey.' arXiv:2407.20018. As of 30 June 2026:
<https://arxiv.org/abs/2407.20018>
- École Polytechnique. 2025. *MGX, Bpifrance, Mistral AI and NVIDIA launch joint venture to build Europe's largest AI campus in France*. Press release, 19 May. As of 6 July 2026:
<https://www.polytechnique.edu/en/press-room/press-releases/mgx-bpifrance-mistral-ai-and-nvidia-launch-joint-venture-build-europes-largest-ai-campus-france>
- EDPB (European Data Protection Board). 2021. *Recommendations 01/2020 on measures that supplement transfer tools to ensure compliance with the EU level of protection of personal data*. Version 2.0, adopted 18 June. As of 6 July 2026:
https://www.edpb.europa.eu/our-work-tools/our-documents/recommendations/recommendations-012020-measures-supplement-transfer_en
- Ember. 2025. 'Grids for data centres: Ambitious grid planning can win Europe's AI race.' 19 June. As of 30 June 2026:
<https://ember-energy.org/latest-insights/grids-for-data-centres-ambitious-grid-planning-can-win-europes-ai-race/>

- ENTSO-E. 2024. *Ten-Year Network Development Plan 2024: Infrastructure Gaps Identification Report*. European Network of Transmission System Operators for Electricity. As of 30 June 2026: <https://www.entsoe.eu/outlooks/tyndp/2024/>
- Epoch AI. 2025a. 'Data on GPU clusters.' As of 30 June 2026: <https://epoch.ai/data/gpu-clusters>
- . 2025b. 'Introducing the Frontier Data Centers Hub.' 4 November. As of 30 June 2026: <https://epoch.ai/blog/introducing-the-frontier-data-centers-hub>
- . 2025c. 'Microsoft's Fairwater datacenter will use more power than Los Angeles.' 26 November. As of 30 June 2026: <https://epoch.ai/data-insights/fairwater-power-usage>
- European Commission. 2021a. 'EU vaccines strategy.' As of 30 June 2026: https://commission.europa.eu/strategy-and-policy/coronavirus-response/public-health/eu-vaccines-strategy_en
- . 2021b. 'Cohesion policy 2021–2027.' As of 30 June 2026: https://ec.europa.eu/regional_policy/2021-2027_en
- . 2022. *REPowerEU Plan*. COM(2022) 230 final. Brussels, 18 May. As of 30 June 2026: <https://eur-lex.europa.eu/legal-content/EN/TXT/?uri=COM:2022:230:FIN>
- . 2025a. 'The Clean Industrial Deal: A joint roadmap for competitiveness and decarbonisation.' COM(2025) 85. Brussels, 26 February. As of 30 June 2026: https://commission.europa.eu/topics/competitiveness/clean-industrial-deal_en
- . 2025b. 'Call for evidence for an impact assessment: EU Cloud and AI Development Act.' Brussels, 9 April. As of 30 June 2026: <https://digital-strategy.ec.europa.eu/en/consultations/have-your-say-future-cloud-and-ai-policies-eu>
- . 2025c. 'Overwhelming response as 76 respondents express interest in the European AI Gigafactories initiative.' 30 June. As of 30 June 2026: <https://digital-strategy.ec.europa.eu/en/news/overwhelming-response-76-respondents-express-interest-european-ai-gigafactories-initiative>
- . 2025d. 'Commission and European Investment Bank Group team up to support AI Gigafactories.' Daily News press release, 4 December. As of 30 June 2026: https://ec.europa.eu/commission/presscorner/detail/en/mex_25_2932
- . 2025e. 'EU launches InvestAI initiative to mobilise €200 billion of investment in artificial intelligence.' Press release IP/25/467, 11 February. As of 30 June 2026: https://ec.europa.eu/commission/presscorner/detail/en/ip_25_467
- . 2026. *Proposal for the Cloud and AI Development Act (CADA)*. Brussels, 3 June. As of 6 July 2026: <https://digital-strategy.ec.europa.eu/en/library/proposal-cloud-and-ai-development-act-cada>

———. n.d. 'AI Gigafactories.' As of 8 July 2026:

https://commission.europa.eu/topics/competitiveness/competitiveness-coordination-tool-projects/ai-gigafactories_en

European Commission & High Representative of the Union for Foreign Affairs and Security Policy.

2025. *Joint Communication to the European Parliament, the Council, the European Economic and Social Committee and the Committee of the Regions on the EU Action Plan on Cable Security.*

JOIN(2025) 9 final. Brussels, 21 February. As of 30 June 2026:

<https://eur-lex.europa.eu/legal-content/EN/TXT/HTML/?uri=CELEX:52025JC0009>

European Data Protection Board. 2021. *Recommendations 01/2020 on measures that supplement transfer tools to ensure compliance with the EU level of protection of personal data.* Version 2.0, 18 June. As of 30 June 2026:

https://www.edpb.europa.eu/system/files/2021-06/edpb_recommendations_202001vo.2.0_supplementarymeasurestransferstools_en.pdf

European Union. 2016. *Regulation (EU) 2016/679 of the European Parliament and of the Council of 27 April 2016 on the protection of natural persons with regard to the processing of personal data and on the free movement of such data, and repealing Directive 95/46/EC (General Data Protection Regulation).* OJ L 119, 1–88. As of 30 June 2026:

<https://eur-lex.europa.eu/eli/reg/2016/679/oj/eng>

Fox, P., M. Schnitzer & D. Privitera. 2025. 'AI compute in Germany: Current capacity, future demand.' KIRA Center, October. As of 30 June 2026:

<https://static1.squarespace.com/static/6560e4863c6eed0712c875f4/t/69132c7891aaf20109e8b9b5/1762864248732/KIRA+Compute+Report+2025+%28English%29.pdf>

Fusion Worldwide. 2025. 'How hyperscaler spending influences semiconductor supply chains.' As of 30 June 2026:

<https://www.fusionww.com/insights/resources/the-cost-of-ai-how-hyperscaler-spending-is-impacting-semiconductor-supply>

Gemini Team. 2023. 'Gemini: A family of highly capable multimodal models.' arXiv:2312.11805. As of 30 June 2026: <https://arxiv.org/abs/2312.11805>

Google DeepMind. 2026. 'Decoupled DiLoCo: Resilient, distributed AI training at scale.' Google DeepMind blog, 23 April. As of 8 July 2026: <https://deepmind.google/blog/decoupled-diloco/>

Goldman Sachs. 2024. *Generational growth: AI, data centers and the coming US power surge.* 28 April. As of 30 June 2026:

<https://www.goldmansachs.com/pdfs/insights/pages/generational-growth-ai-data-centers-and-the-coming-us-power-surge/report.pdf>

GridLab. 2025. *Practical guidance and considerations for large load interconnections.* March. As of 30 June 2026:

<https://gridlab.org/wp-content/uploads/2025/03/GridLab-Report-Large-Loads-Interim-Report.pdf>

- Guardian. 2025. 'Zuckerberg says Meta will build data center the size of Manhattan in latest AI push.' 16 July. As of 30 June 2026: <https://www.theguardian.com/technology/2025/jul/16/zuckerberg-meta-data-center-ai-manhattan>
- Hochreiter, S., & J. Schmidhuber. 1997. 'Long short-term memory.' *Neural Computation* 9 (8): 1735–80. As of 30 June 2026: <https://doi.org/10.1162/neco.1997.9.8.1735>
- IEA (International Energy Agency). 2022. *Norway 2022 energy policy review*. IEA Energy Policy Reviews. OECD Publishing, Paris. As of 30 June 2026: <https://doi.org/10.1787/7411c642-en>
- . 2025. *Energy and AI*. International Energy Agency, Paris. As of 30 June 2026: <https://www.iea.org/reports/energy-and-ai>
- IEEE ComSoc Technology Blog. 2025. 'Hyperscaler capex > \$600 bn in 2026 a 36% increase over 2025 while global spending on cloud infrastructure services skyrockets.' IEEE Communications Society, 22 December. As of 30 June 2026: <https://techblog.comsoc.org/2025/12/22/hyperscaler-capex-600-bn-in-2026-a-36-increase-over-2025-while-global-spending-on-cloud-infrastructure-services-skyrockets/>
- Jaghoular, S., J.M. Ong & J. Hagemann. 2024a. 'OpenDiLoCo: An open-source framework for globally distributed low-communication training.' arXiv:2407.07852. As of 30 June 2026: <https://arxiv.org/abs/2407.07852>
- Jaghoular, S., J.M. Ong, M. Basra, F. Obeid, J. Straube, M. Keiblinger, et al. 2024b. 'INTELLECT-1 technical report.' arXiv: 2412.01152. As of 30 June 2026: <https://arxiv.org/abs/2412.01152>
- Jacamon, V., J. Dallard & T. Spencer. 2025. 'Overcoming energy constraints is key to delivering on Europe's data centre goals.' International Energy Agency, Paris, 16 November. As of 30 June 2026: <https://www.iea.org/commentaries/overcoming-energy-constraints-is-key-to-delivering-on-europe-s-data-centre-goals>
- Kale, S., A. Douillard & Y. Donchev. 2025. 'Eager updates for overlapped communication and computation in DiLoCo.' arXiv:2502.12996. As of 30 June 2026: <https://arxiv.org/abs/2502.12996>
- Kirpsza, A. 2022. 'Time is of the essence: Explaining the duration of European Union lawmaking under the co-decision procedure.' *European Union Politics* 23 (1): 22–43. As of 30 June 2026: <https://doi.org/10.1177/14651165211036423>
- Kothuri, A., D. Janků, J.M. Teixeira & D. Juijn. 2025. 'Submission to the call for evidence on the Cloud and AI Development Act.' Centre for Future Generations, 16 July. As of 30 June 2026: <https://cfg.eu/special-compute-zones-submission/>
- Kryś, J., Y. Sharma & J. Egan. 2025. 'Distributed and decentralised training: Technical governance challenges in a shifting AI landscape.' arXiv:2507.07765. Accepted as an oral presentation at the Technical AI Governance Workshop, ICML 2025. As of 30 June 2026: <https://arxiv.org/abs/2507.07765>

- Lidin, J., A. Sarfi, E. Miahi, Q. Anthony, S. Chauhan, E. Pappas, et al. 2026. 'Covenant-72B: Pre-training a 72B LLM with trustless peers over-the-internet.' arXiv:2603.08163. As of 8 July 2026: <https://arxiv.org/abs/2603.08163>
- Lindner, J., & S. Mack. 2024. 'Capital Markets Union: Europe must stop beating around the bush.' Policy Brief, Jacques Delors Centre, 11 July. As of 30 June 2026: <https://www.delorscentre.eu/en/publications/detail/publication/capital-markets-union>
- McKinsey & Company. 2025. 'The cost of compute: A \$7 trillion race to scale data centers.' 28 April. As of 30 June 2026: <https://www.mckinsey.com/industries/technology-media-and-telecommunications/our-insights/the-cost-of-compute-a-7-trillion-dollar-race-to-scale-data-centers>
- McMahan, H.B., E. Moore, D. Ramage, S. Hampson & B.A. y Arcas. 2017. 'Communication-efficient learning of deep networks from decentralized data.' *Proceedings of the 20th International Conference on Artificial Intelligence and Statistics (AISTATS)*, JMLR 54:1273–82. As of 30 June 2026: <https://arxiv.org/abs/1602.05629>
- Mistral AI. 2025. 'Mistral AI raises €1.7B to accelerate technological progress with AI.' Press release, 9 September. As of 30 June 2026: <https://mistral.ai/news/mistral-ai-raises-1-7-b-to-accelerate-technological-progress-with-ai>
- Morris, A. 2025. 'EU defers formal call for AI gigafactories to early 2026.' Light Reading, 4 December. As of 30 June 2026: <https://www.lightreading.com/ai-machine-learning/eu-defers-formal-call-for-ai-gigafactories-to-early-2026>
- NERC (North American Electric Reliability Corporation). 2025. *2025 Summer Reliability Assessment*. May. As of 30 June 2026: https://www.nerc.com/globalassets/programs/rapa/ra/nerc_sra_2025.pdf
- Nicoli, F., & R. Beetsma. 2024. 'Joint public procurement as a tool for European Union industrial policy.' Bruegel, Policy Brief 18/2024. As of 30 June 2026: <https://www.bruegel.org/policy-brief/joint-public-procurement-tool-european-union-industrial-policy>
- Norris, T.H., T. Profeta, D. Patiño-Echeverri & A. Cowie-Haskell. 2025. 'Rethinking load growth: Assessing the potential for integration of large flexible loads in US power systems.' NI R 25-01. Nicholas Institute for Energy, Environment & Sustainability, Duke University, February. As of 30 June 2026: <https://nicholasinstitute.duke.edu/publications/rethinking-load-growth>
- Nscale, Aker & OpenAI. 2025. 'Nscale, Aker and OpenAI to establish Stargate Norway: A 100,000 NVIDIA GPU AI Gigafactory powered by renewable energy in Northern Norway.' Press release, 31 July. As of 30 June 2026: <https://www.nscale.com/press-releases/stargate-norway-nscale-aker-openai>

- NVIDIA. 2024. 'NVIDIA Blackwell platform arrives to power a new era of computing.' Press release, 18 March. As of 30 June 2026:
<https://nvidianews.nvidia.com/news/nvidia-blackwell-platform-arrives-to-power-a-new-era-of-computing>
- OECD (Organisation for Economic Co-operation and Development). 2026. 'Venture capital investments in artificial intelligence through 2025.' OECD Publishing. As of 30 June 2026:
https://www.oecd.org/en/publications/venture-capital-investments-in-artificial-intelligence-through-2025_a13752f5-en.html
- OpenAI. 2024. 'OpenAI o1 system card.' 5 December. As of 30 June 2026:
<https://cdn.openai.com/o1-system-card-20241205.pdf>
- . 2025a. 'GPT-4.5 system card.' 27 February. As of 30 June 2026:
<https://cdn.openai.com/gpt-4-5-system-card-2272025.pdf>
- . 2025b. 'Introducing Stargate Norway.' 31 July. As of 30 June 2026:
<https://openai.com/index/introducing-stargate-norway/>
- . 2025c. 'OpenAI, Oracle, and SoftBank expand Stargate with five new AI data center sites.' 23 September. As of 30 June 2026: <https://openai.com/index/five-new-stargate-sites/>
- Orrick, Herrington & Sutcliffe LLP. 2025. 'Megawatts to megabytes: Orrick's 2025 guide to developing, financing & powering data centers.' 15 July. As of 30 June 2026:
<https://www.orrick.com/en/Insights/2025/07/Megawatts-to-Megabytes-Orricks-2025-Guide-to-Developing-Financing-and-Powering-Data-Centers>
- Patel, D., D. Nishball & J. Eliahou Ontiveros. 2024a. 'AI datacenter energy dilemma: Race for AI datacenter space.' SemiAnalysis, 13 March. As of 30 June 2026:
<https://semianalysis.com/2024/03/13/ai-datacenter-energy-dilemma-race/>
- . 2024b. 'Multi-Datacenter Training: OpenAI's ambitious plan to beat Google's infrastructure.' SemiAnalysis, 4 September. As of 30 June 2026:
<https://semianalysis.com/2024/09/04/multi-datacenter-training-openais/>
- Phillips-Robins, A., & S. Winter-Levy. 2025. 'How to gauge whether Trump's AI chip deals with Gulf countries are any good.' Carnegie Endowment for International Peace, 16 May. As of 30 June 2026:
<https://carnegieendowment.org/emissary/2025/05/ai-chip-trump-gulf-uae-saudi-security-risk-good-deal>
- Pilz, K.F., Y. Mahmood & L. Heim. 2025. *AI's power requirements under exponential growth: Extrapolating AI data center power demand and assessing its potential impact on U.S. competitiveness*. Santa Monica, Calif.: RAND Corporation. RR-A3572-1. As of 30 June 2026:
https://www.rand.org/pubs/research_reports/RRA3572-1.html
- Pilz, K.F., J. Sanders, R. Rahman & L. Heim. 2025. 'Trends in AI supercomputers.' arXiv:2504.16026. As of 30 June 2026: <https://arxiv.org/abs/2504.16026>

- Prime Intellect. 2024. 'OpenDiLoCo: First truly distributed training across continents.' 11 July. As of 30 June 2026: <https://www.primeintellect.ai/blog/opensdiloco>
- Pure Storage. 2024. 'What is a hyperscale data center?' As of 30 June 2026: <https://www.purestorage.com/knowledge/what-is-a-hyperscale-data-center.html>
- Qi, J., W. Zhu, L. Li, M. Wu, Y. Wu, W. He, et al. 2025. 'DiLoCoX: A low-communication large-scale training framework for decentralized cluster.' arXiv:2506.21263. As of 30 June 2026: <https://arxiv.org/abs/2506.21263>
- Reuters. 2025a. 'Musk's xAI increased Tennessee gas turbines without permits, community groups say.' 9 April. As of 30 June 2026: <https://www.reuters.com/sustainability/boards-policy-regulation/musks-xai-increased-tennessee-gas-turbines-without-permits-community-groups-say-2025-04-09/>
- . 2025b. 'Europe's AI gigafactory push attracts 76 bids, EU tech chief says.' 30 June. As of 30 June 2026: <https://www.reuters.com/sustainability/boards-policy-regulation/europes-ai-gigafactory-push-attracts-76-bids-eu-tech-chief-says-2025-06-30/>
- S&P Global. 2021. 'French power demand prices rise on cold spell.' 9 February. As of 30 June 2026: <https://www.spglobal.com/commodity-insights/en/news-research/latest-news/electric-power/020921-french-power-demand-prices-rise-on-cold-spell>
- Sevilla, J., T. Besiroglu, B. Cottier, J. You, E. Roldán, P. Villalobos & E. Erdil. 2024. 'Can AI scaling continue through 2030?' Epoch AI, 20 August. As of 30 June 2026: <https://epoch.ai/blog/can-ai-scaling-continue-through-2030>
- Sevilla, J., & A. Troynikov. 2025. 'Could decentralized training solve AI's power problem?' Epoch AI, 28 October. As of 30 June 2026: <https://epoch.ai/blog/could-decentralized-training-solve-ais-power-problem>
- Shehabi, A., et al. 2024. *2024 United States data center energy usage report*. LBNL-2001637. Lawrence Berkeley National Laboratory. As of 30 June 2026: https://eta-publications.lbl.gov/sites/default/files/2024-12/lbnl-2024-united-states-data-center-energy-usage-report_1.pdf
- Smith, M.S. 2026. 'What will it take to build the world's largest data center?' IEEE Spectrum, 24 March. As of 30 June 2026: <https://spectrum.ieee.org/5gw-data-center>
- Stanford HAI (Stanford Institute for Human-Centered Artificial Intelligence). 2025. *Artificial Intelligence Index Report 2025*. Stanford University. As of 30 June 2026: <https://hai.stanford.edu/ai-index/2025-ai-index-report>
- Tagliapietra, S. 2026. 'How will the Iran conflict hit European energy markets?' Bruegel First Glance, 2 March. As of 30 June 2026: <https://www.bruegel.org/first-glance/how-will-iran-conflict-hit-european-energy-markets>

- Tanner, B., D. Belle, C.F. Kerry, N. Kyosovska, A. Renda, E. Tabassi & A.W. Wyckoff. 2026. 'Global energy demands within the AI regulatory landscape.' Brookings Institution, 10 April. As of 30 June 2026:
<https://www.brookings.edu/articles/global-energy-demands-within-the-ai-regulatory-landscape/>
- Turing, A.M. 1950. 'Computing machinery and intelligence.' *Mind* 59 (236): 433–60.
- White House. 2025. 'Accelerating federal permitting of data center infrastructure.' 23 July. As of 30 June 2026:
<https://www.whitehouse.gov/presidential-actions/2025/07/accelerating-federal-permitting-of-data-center-infrastructure/>
- Zhu, L., Z. Liu & S. Han. 2019. 'Deep leakage from gradients.' *Advances in Neural Information Processing Systems* 32: 14747–56). As of 30 June 2026: <https://arxiv.org/abs/1906.08935>

Annex A. Converting between H100 equivalents and power capacity

This annex sets out the rough estimates used to convert our suggested 10–20 million H100 equivalents to total facility power. They are based on two recently published estimates that turn out to converge closely once their underlying assumptions are unpacked.

A.1. Source 1: RAND (Pilz, Mahmood & Heim 2025)

RAND reports a power draw of 1,419 W per H100 equivalent at the cluster level – that is, the GPU together with supporting hardware including server, networking and other in-rack components, but excluding datacentre power usage effectiveness (PUE) overhead such as cooling and power conversion. A PUE of 1.25 means the facility consumes 25 per cent more power than the IT hardware draws on its own. This figure is taken from NVIDIA’s DGX H100 SuperPOD reference design. To obtain total facility power, RAND separately applies a PUE multiplier, using 1.25 as a near-term value and projecting improvement to 1.05 by 2030.

A.2. Source 2: Epoch AI (Sevilla et al. 2024)

Epoch AI reports a power draw of 1,700 W per H100 equivalent at the total facility level. Their derivation is explicit: a 700 W thermal design power (TDP), the chip’s nominal sustained power draw, for the H100, multiplied by a 1.96× cluster overhead factor (based on the DGX H100 SuperPOD server expected average power), and then by a PUE of 1.2. Backing the PUE out yields approximately 1,417 W at the cluster level, nearly identical to RAND’s 1,419 W.

A.3. Averaged estimate

The two sources converge on approximately 1,418 W per H100 equivalent at the cluster level. The remaining difference between them reflects the choice of PUE multiplier rather than any disagreement about the underlying hardware. As detailed in Table A.1, applying the central PUE assumption of 1.2 yields a facility-level figure of approximately 1,700 W per H100 equivalent, which we adopt as our central estimate; for sensitivity, we also report figures under PUE = 1.25 (near-term) and PUE = 1.1 (improved efficiency by 2030).

Table A.1. Total facility power required, by compute target and PUE assumption

Compute target	Central (PUE 1.2)	Near-term (PUE 1.25)	Improved (PUE 1.1)
10M H100e	~17.0 GW	~17.7 GW	~15.6 GW
15M H100e	~25.5 GW	~26.6 GW	~23.4 GW
20M H100e	~34.0 GW	~35.5 GW	~31.2 GW

Source: Authors' calculation based on Pilz, Mahmood & Heim (2025) and Sevilla et al. (2024).

A.4. Caveats

This conversion is subject to several factors including utilisation rates, cooling technology, and next-generation power-efficient chips (NVIDIA Blackwell delivers roughly 35 per cent greater power efficiency than the H100, with Rubin expected to deliver a comparable improvement). By the time European infrastructure at this scale is operational, per-compute power requirements could be meaningfully lower. On the other hand, demand-side estimates of compute required for frontier training in 2030 also rest on multiple assumptions, and could be higher than currently projected. The figures given above should be read as order-of-magnitude working estimates rather than precise forecasts.